Performative Prediction in a Single-Asset Market

Self-Fulfilment and Signal Erosion under Forecast Adoption

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Declaration on the use of AI. Generative AI tools, including large language models, were used in the course of this project to assist with software implementation, debugging, figure generation, and language editing. The research questions, model design, experimental design, analysis, interpretation of results, and conclusions are the author's own. The author has reviewed all AI-assisted output and takes full responsibility for the content of this report.

Abstract

Financial forecasts are usually evaluated as if they were passive descriptions of an external market process. In real markets, however, widely adopted forecasts can change the outcomes they are meant to predict: traders act on the signal, their coordinated orders move prices, and the realised return used to score the forecast may partly reflect the forecast's own impact. This creates a measurement problem. When forecast performance changes after adoption, it is unclear whether the model has become more informative, whether the market has changed, or whether the evaluation target has been contaminated by self-fulfilling price impact.

This report studies that problem in a minimal single-asset agent-based market combining Brock-Hommes mean-variance demand with a Beja-Goldman / Farmer-Joshi market maker. Traders switch from a no-skill null rule to a shared autoregressive return forecast through either exogenous diffusion or endogenous certainty-equivalent switching. The same forecast is then evaluated on the same simulated paths against two targets: the realised return observed by a deployed forecaster, and a demand-adjusted counterfactual that removes contemporaneous price impact.

The results reveal a dual-channel effect. As adoption increases, coordinated forecast-driven demand raises the realised-return out-of-sample R^2 from 0.07 under a zero-adoption control to roughly 0.19 near full adoption. At the same time, the demand-adjusted R^2 falls from 0.08 to 0.04, while the effective AR coefficient of realised returns rises from 0.28 to 0.45. Adoption thus improves apparent predictive performance against realised returns through a self-fulfilling price-impact channel, while reducing performance against the demand-adjusted component the forecast was designed to capture. The central implication is that, once a forecast is widely adopted, standard realised-return evaluation can conflate genuine predictive content with the market impact generated by the forecast's own users.

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1. Introduction

1.1 Motivation

Consider a forecasting service that publishes a one-period-ahead return forecast for a single asset. At launch nobody trades on it, and its out-of-sample accuracy measures whatever genuine structure the rule has found. Then it acquires users: subscribers trade in the direction of the published signal, their coordinated orders move the price, and the returns against which the forecast is scored begin to carry its fingerprint. When the tracked performance metrics now shift, what exactly has changed: the forecast's skill, the market, or the meaning of the measurement?

Both readings are defensible. The improvement could be real, since coordinated trading after a positive forecast pushes the realised return in the forecast's direction, so the forecast points where the price actually goes. It could equally be misleading: the genuine, independent pattern the forecast was built to detect may be eroding, even as the very adoption that inflates its measured score hides that decline. This report builds the smallest market in which that question can be posed, and answered, exactly.

1.2 Relation to previous work

Three research traditions meet in this question. The first is reflexivity and its machine-learning formalisation, performative prediction: Soros [1987] argued that market participants' views alter the outcomes those views are about, and Perdomo et al. [2020] made the deployed model's effect on its own evaluation distribution a formal object of study. The second is heterogeneous-agent finance, which supplies the market: mean-variance traders who switch between competing rules on recent performance, after Brock and Hommes [1998], trading against a market maker who moves the price linearly in net order flow, in the disequilibrium tradition of Beja and Goldman [1980] and Farmer and Joshi [2002]. The third is classical out-of-sample forecast evaluation in the tradition of Diebold and Mariano [1995], whose machinery presumes a target that the forecast under test does not move, exactly the premise adoption breaks. This report brings the three traditions together. It sets a performative forecast inside a heterogeneous-agent market and scores it with standard out-of-sample metrics. The divergence between the two evaluation targets is its central result.

1.3 Research question and contribution

The research question is: how does adoption of a shared autoregressive return forecast affect its realised-return predictive performance (its accuracy against the return the market actually prints) and its demand-adjusted predictive performance (its accuracy against a counterfactual return that strips out the forecast's own contemporaneous price impact), in a tractable single-asset agent-based market? Both readings score the same forecast on the same path; the realised return is the only one a deployed forecaster can actually observe, while the demand-adjusted return is a counterfactual available only inside the model.

The contribution is a small, fully reproducible simulator that demonstrates and quantifies a dual-channel answer. At the baseline configuration, adoption lifts the realised-return out-of-sample R^2 from 0.07 under a zero-adoption control to 0.19 at near-full adoption, while the demand-adjusted reading falls from 0.08 to 0.04 and the effective AR coefficient rises from 0.28 to 0.45 (section 5.3). A 3600-run sweep then maps both channels across price impact μ, input persistence φ, estimation window w, and AR order p, locating the adoption shares at which the independent signal has lost half its low-adoption baseline (section 5.5).

1.4 Report roadmap

The remainder of the report proceeds as follows. Section 2 develops the theoretical background and fixes notation. Section 3 specifies the market, the trader layers, the forecasting rules, the adoption mechanisms, and the realised versus demand-adjusted decomposition. Section 4 defines the evaluation endpoints and the experimental design. Section 5 reports the results, from the baseline market to the parameter sweep and critical adoption shares. Section 6 interprets the mechanism, states its boundary conditions, and lists limitations, and section 7 concludes. Appendix A describes the software and its phased construction.

2. Background and theoretical framework

2.1 Reflexivity and performative prediction

The report rests on two framing ideas; the older is reflexivity. Soros [1987] argued that financial markets are not passive measurement devices: participants act on their views of the market, and through those actions alter the prices, and at times the fundamentals, that the views are about. Expectation and outcome are linked in both directions, so a view can come true partly because it is widely held, the self-fulfilling case, or undo itself by changing the conditions that once made it accurate.

Machine learning reached the same structure from the deployment side. Perdomo et al. [2020] formalise performative prediction: prediction problems in which the deployed model shifts the distribution of the data on which it is judged, so that measured accuracy is a property of the model and the behavioural response together, not of the model alone. They draw a key distinction between two kinds of model. One is merely a fixed point of retraining, having settled into consistency with the data its own use induces; the other stays genuinely optimal even once that induced shift is accounted for. Naming this difference made the deployment effect an object of study in its own right.

The two framings describe the same situation from different sides. A public return forecast that acquires users with price impact is reflexive in the first sense and performative in the second, since the orders it coordinates move the market it describes. Once that happens, scoring the forecast with an off-the-shelf accuracy metric no longer measures forecast skill alone; the score mixes skill with whatever part of the outcome the forecast's own users produced, and the two parts need not move in the same direction. The market of section 3 makes the feedback channel a single explicit term, so the mixture can be taken apart exactly.

2.2 Heterogeneous-agent finance and price formation

The demand side of the model comes from heterogeneous-agent finance, in which market dynamics are generated by interacting traders using different, competing rules. The reference framework is Brock and Hommes [1998]: each trader is a myopic mean-variance maximiser with constant absolute risk aversion, holding a position proportional to the expected next-period return and inversely proportional to risk aversion times perceived variance, and traders switch between forecasting rules through a discrete-choice mechanism driven by recent relative performance. Coupling rule choice to rule performance this way can destabilise an otherwise simple asset market, the routes to chaos of their title. The present design takes from the framework the zero-dividend special case of the demand function and the binary discrete-choice switch, applied to one shared forecasting rule against a no-skill null (section 3.2).

Price formation comes from the disequilibrium market-adjustment tradition. Beja and Goldman [1980] study prices that respond to excess demand at a finite rate instead of clearing every period, so trading pressure moves the quote directly, and show that introducing trend-chasing speculation into such a market can destabilise it. Farmer and Joshi [2002] cast the same idea in discrete time as a market maker who absorbs net order flow and shifts the log-price linearly in it, and show that common trading strategies leave statistical signatures in returns, trend following in particular inducing positive short-horizon autocorrelation. Both results carry weight here: linear impact is the only channel through which forecast use feeds back into returns, and the induced autocorrelation licenses the small reduced-form AR component the shared rule is built to recover (section 3.1).

2.3 Forecast evaluation under self-fulfilment

Out-of-sample forecast evaluation has a settled routine: fit on data through period t, forecast period $t+1$, accumulate mean squared forecast error, and summarise the result against a benchmark as an out-of-sample R^2. The routine's unstated premise is that the target is exogenous of the forecast, meaning the return being predicted would have been the same had the forecast never been issued, so the score measures the forecast and nothing else.

Adoption removes that premise. Once traders act on a public forecast, their coordinated orders enter the realised return through price impact, so the target partially inherits the forecast it is supposed to test. A single off-the-shelf score then mixes two channels, the independent signal the rule was estimated to capture and the self-fulfilment its users inject, and because the two carry opposite signs in this setting, the score can improve while the independent content behind it decays. No refinement of estimation windows, benchmarks, or test statistics repairs this, because the defect sits in the target, not in the procedure.

The response of this design is a decomposition rather than a new metric. The same forecast is scored on the same path against two targets, the realised return and a demand-adjusted counterfactual with contemporaneous price impact stripped out, and the wedge between the two readings is attributed to deployment rather than skill. Section 4 defines the resulting endpoints exactly; section 6.1 reads the simulated findings back through this lens.

2.4 Key concepts and notation

Four terms carry the argument. The *adoption share* is the fraction of traders currently trading on the public forecast, moved either by exogenous diffusion at rate π or by endogenous performance-based switching (section 3.4). The *realised return* is the return the market actually prints, contemporaneous price impact included: the only target a deployed forecaster can observe. The *demand-adjusted return* $x_{t+1} = r_{t+1} - \mu D_t$ is its counterfactual companion, removing the contemporaneous price-impact term only while leaving the lagged-return component in place; it is not the full residual of a return regression, and it asks what the forecast would have scored had its users not moved the current quote (section 3.6). The *effective AR coefficient* is the lag-1 autocorrelation of realised returns in a trailing window: it sits at the input φ without adoption, rises above it when coordinated adopter demand reinforces the lagged-return component, and is the diagnostic connecting the two R^2 readings (section 4.2). The symbols that recur throughout are gathered for reference in Table 1 at the end of section 3.

3. The model

3.1 Single-asset market with linear price impact

The market is deliberately minimal. Time is discrete, $t = 0, 1, \dots, T$. A single risky asset trades at log-price p_t with one-period return $r_{t+1} = p_{t+1} - p_t$; the risk-free rate is normalised to zero and the asset pays no dividend, so r_{t+1} is an excess return. Each of N traders submits a signed order $q_{i,t}$ every period. There is no Walrasian auctioneer (the fictitious agent of general-equilibrium theory who, before any trade takes place, finds the single price at which supply and demand exactly balance, so the market always clears instantaneously): following the disequilibrium market-adjustment tradition, a risk-neutral market maker absorbs whatever aggregate order flow arrives and shifts the log-price in proportion to it [Beja and Goldman, 1980, Farmer and Joshi, 2002]. Linear price impact is thus the single channel through which trading feeds back into returns, the carrier of every reflexive effect this report studies.

Aggregate signed demand is the per-trader average order,

$$D_t \;=\; \frac{1}{N}\sum_{i=1}^{N} q_{i,t}, \tag{1}$$

and the market maker updates the log-price according to

$$p_{t+1} \;=\; p_t \;+\; \mu\, D_t \;+\; \sigma\, \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim \mathcal{N}(0,1), \tag{2}$$

where the price-impact coefficient $\mu = 1/\lambda > 0$ is the reciprocal of the market maker's liquidity λ and $\sigma > 0$ scales the exogenous news shock; (2) is the zero-fundamental special case of the Farmer-Joshi log-linear price update. First-differencing (2) gives a one-period market-maker return of $\mu D_t + \sigma\varepsilon_{t+1}$; the baseline return law adds an autoregressive component carried in from prior periods,

$$r_{t+1} \;=\; \varphi\, r_t \;+\; \mu\, D_t \;+\; \sigma\, \varepsilon_{t+1}, \qquad |\varphi| < 1 \text{ small.} \tag{3}$$

Equation (3) splits the return into the three pieces the model works with: a small predictable component φr_t, the signal the rolling AR rule of section 3.3 estimates; the contemporaneous price impact μD_t, the only term through which trader behaviour enters; and unpredictable news $\sigma\varepsilon_{t+1}$. The φr_t term is not ad hoc: Farmer and Joshi [2002] show that linear trend-following demand induces positive first-lag return autocorrelation of the same order as its intensity, which (3) absorbs as a single reduced-form coefficient rather than a separate trader type.

One modelling choice deserves note. Where Farmer and Joshi [2002] have the market maker absorb net order flow (changes in positions), the present market maker absorbs the level of aggregate demand each period, closer to the Beja-Goldman excess-demand adjustment, so a persistent position exerts persistent price pressure.

3.2 Trader behaviour: belief, decision, and switching layers

Trader behaviour decomposes into three layers, kept separate so the results can say where each channel of the dual-channel result is produced: belief (what return the agent expects), decision (how that belief becomes a position), and switching (whether the agent uses the shared rule at all). The mechanism under study operates on the belief layer, where the rolling AR forecast is public information, and reaches returns only through the decision and switching layers, where action is private. The belief layer has exactly two states: an adopter's belief is the shared rolling autoregressive forecast $\hat{r}^{(A)}_{t+1}$ of section 3.3, identical across adopters because the rule is public, while a non-adopter holds no belief about returns at all, the null rule of the same section generating order flow that is directionally uninformative on average, an active but skill-free benchmark rather than a behavioural model.

The decision layer follows Brock and Hommes [1998]: each trader is a myopic mean-variance maximiser with constant absolute risk aversion. With conditional forecast $\hat{r}_{i,t+1}$ of the next-period return and common perceived conditional variance $\sigma^2 > 0$, the optimal signed position is

$$z_{i,t} \;=\; \frac{\hat{r}_{i,t+1}}{a\,\sigma^2}, \tag{4}$$

where $a > 0$ is the coefficient of absolute risk aversion; this is the zero-dividend, zero-rate special case of the Brock-Hommes demand function: buy on a positive expected return, sell on a negative one, trade smaller when the market is perceived as riskier. Adopters evaluate (4) at the public forecast, with the result clipped at a position cap,

$$q^{(A)}_{i,t} \;=\; \mathrm{clip}\left(\frac{\hat{r}^{(A)}_{t+1}}{a\,\sigma^2},\; -\bar{q}_i,\; +\bar{q}_i\right), \tag{5}$$

where the cap $\bar{q}_i > 0$ is a leverage or inventory constraint: rolling-window estimates of $\hat{r}^{(A)}_{t+1}$ are noisy, and the cap prevents a single extreme forecast from generating an unboundedly large trade. Non-adopters bypass the demand mapping entirely; their null order enters as drawn.

The switching layer is a single binary state: $a_{i,t} \in \{0,1\}$ indicates whether trader i has adopted the shared rule at time t. The final order is $q_{i,t} = (1-a_{i,t})\,q^{(0)}_{i,t} + a_{i,t}\,q^{(A)}_{i,t}$, and the one-period profit feeding the performance-based switching rule is $\Pi_{i,t+1} = q_{i,t}\,r_{t+1}$; how $a_{i,t}$ evolves, by stochastic diffusion or certainty-equivalent comparison, is the subject of section 3.4. Two details matter later. First, only the product $a\sigma^2$ enters the demand mapping, so it is treated as a single risk-scale parameter. Second, the advanced order returns zero whenever the forecast is not finite, so the warm-up periods before the first rolling window fills leave adopters inactive rather than injecting undefined demand.

3.3 Null rule and rolling AR forecasting rule

Non-adopters trade by the null rule: trader i submits $q^{(0)}_{i,t} = \xi_{i,t}$ with $\xi_{i,t} \sim \mathcal{N}(0,\sigma_q^2)$, drawn independently across traders and time. The rule is a reference point, not a model of real trading: it keeps every trader active while injecting zero-mean, directionally uninformative order flow, so any systematic predictability the advanced rule generates is measured against a market in which trading happens but is not forecast-driven. One design feature is worth noting: the null order bypasses the demand mapping and carries no position cap, entering aggregate demand as drawn with typical magnitude set by σ_q alone, while the advanced order (5) is clipped at $\bar{q}_i$.

Adopters share a single advanced rule: an autoregressive forecast of the next return, estimated on a rolling window of recent returns. With rolling-window length w and autoregressive order p, at time t the public model fits by least squares

$$r_{s+1} \;=\; c \;+\; \sum_{j=1}^{p} \varphi_j\, r_{s+1-j} \;+\; u_{s+1}, \tag{6}$$

using only the observations in the window $s = t-w,\dots,t-1$, and issues the one-step-ahead out-of-sample forecast

$$\hat{r}^{(A)}_{t+1}(w,p) \;=\; \hat{c}_{t,w,p} \;+\; \sum_{j=1}^{p} \hat{\varphi}_{j,t,w,p}\, r_{t+1-j}. \tag{7}$$

Nothing estimated at time t uses information from after t, so the forecast is out of sample by construction; rolling estimation lets the coefficients adapt when the return process changes, which under adoption it does, the fit of (6) being the object that later absorbs the amplified empirical autocorrelation of realised returns. The rule is built to recover the φr_t component of (3), the simplest forecasting environment that should in principle work; penalised regressions, tree ensembles, and recurrent architectures are deliberately excluded. The fit is ordinary least squares on a lagged design with intercept, defaulting to $p = 1$; the first w periods, before the rolling window fills, produce no forecast and leave adopters inactive. The core design holds $p = 1$ with windows $w \in \{50, 100, 250\}$; higher orders $p \in \{2, 5, 10\}$ enter only as a phase 6 robustness check.

3.4 Adoption mechanisms

Two mechanisms govern how the adoption indicator $a_{i,t}$ evolves; both act on the switching layer alone, and the simulator runs exactly one at a time. The stochastic mechanism, easier to interpret, asks whether wider use of the forecast is by itself enough to change predictability; the performance-based mechanism studies the feedback between early success and later crowding.

3.4.1 Stochastic diffusion

Under stochastic diffusion each non-adopter discovers and adopts the public rule with a fixed probability $\pi \in (0, 1)$ per period, and each adopter abandons it with an optional small exit probability δ:

$$P(a_{i,t+1} = 1 \mid a_{i,t} = 0) \;=\; \pi, \tag{8}$$

$$P(a_{i,t+1} = 0 \mid a_{i,t} = 1) \;=\; \delta. \tag{9}$$

The draws are independent across agents and periods, so the adoption share $A_t = (1/N)\sum_i a_{i,t}$ moves at an expected per-period rate of $\pi(1 - A_t) - \delta A_t$, with no social learning and no profitability comparison: adoption is imposed on the market rather than generated by it, isolating the pure effect of wider use of a common signal. One implementation detail matters for every later comparison: adoption draws come from a dedicated random generator, so changing π or δ cannot advance the market shock stream, and regimes with different adoption settings run on paired shocks.

3.4.2 *Certainty-equivalent performance-based switching*

The economically meaningful mechanism makes entry depend on recent relative risk-adjusted success, scored in the same mean-variance terms as the decision layer. Each rule $k \in \{(0),(A)\}$ carries a one-period certainty-equivalent return,

$$\mathrm{CE}_t^{(k)} \;=\; \bar{\Pi}_t^{(k)} \;-\; \tfrac{a}{2}\,\hat{\sigma}_t^{2,(k)}, \tag{10}$$

where $\bar{\Pi}_t^{(k)}$ and $\hat{\sigma}_t^{2,(k)}$ are the rolling mean and variance of the rule's per-period profit over a window of length W_A, and a is the risk-aversion coefficient of the decision layer. The switching score is the gap $S_t = \mathrm{CE}_t^{(A)} - \mathrm{CE}_t^{(0)}$, and each non-adopter switches to the advanced rule with the logistic probability

$$P(a_{i,t+1} = 1 \mid a_{i,t} = 0) \;=\; \Lambda(\alpha + \beta\, S_t), \qquad \Lambda(z) = \frac{1}{1+e^{-z}}, \tag{11}$$

where α is a baseline adoption term and $\beta > 0$ controls sensitivity to recent performance; the specification is the binary Brock and Hommes [1998] discrete-choice mechanism applied at the switching layer. Two further choices complete the rule. First, the design specifies an entry probability but no exit analogue of (9), and the model keeps that asymmetry, so adopters retain their state and the share can only ratchet upward. Second, the profit streams feeding (10) belong to one representative trader per rule, the common capped adopter order and the realised order of a single representative non-adopter, keeping the per-trader mean and variance of null profit independent of the adoption share. Unlike stochastic diffusion, this mechanism closes a second reflexive loop: early adopter success raises $\mathrm{CE}_t^{(A)}$ and accelerates entry, and the entrants' own trading reshapes the returns from which every later certainty equivalent is computed.

3.5 Intra-period timing

The preceding equations fix what each trader and the market maker do, but not when within a period they do it. The positive-sign impact term μD_t in (3) makes that order of operations load-bearing: adopter demand formed on today's forecast moves today's quote, so forecast and evaluation target are entangled within the period, and self-fulfilment cannot be told from erosion unless the sequence of events is explicit. The design fixes the following sequence inside each period, and the simulator follows it exactly.

1. Agents observe returns up to and including period t and form forecasts $\hat{r}_{t+1}^{(A)}$ per (7).
2. Adopters and non-adopters submit orders $q_{i,t}$: the advanced order (5) and the null order, combined through the adoption indicator $a_{i,t}$ as in section 3.2.
3. The market maker absorbs aggregate demand D_t and moves the quote immediately, as in the price update (2).

4. The exogenous news shock $\sigma\,\varepsilon_{t+1}$ and the autoregressive term $\varphi\, r_t$ realise.

The sequence separates what the forecast knows from what its users cause: returns through period t are in hand before any order is placed, and the μD_t move of step 3 is in the books before next period's news arrives in step 4, making it well defined to score the same forecast against two targets on the same path, the realised return including the step 3 impact or a counterfactual stripping it out (next subsection).

3.6 The realised vs demand-adjusted decomposition

For evaluation the return law (3) is rewritten as a contemporaneous price-impact component plus an ex-feedback remainder: the realised return splits as

$$r_{t+1} \;=\; \mu\, D_t \;+\; x_{t+1}, \tag{12}$$

where

$$x_{t+1} \;=\; r_{t+1} \;-\; \mu\, D_t \;=\; \varphi\, r_t \;+\; \sigma\,\varepsilon_{t+1} \tag{13}$$

is the demand-adjusted return, the part of the period-$(t+1)$ return that does not depend on contemporaneous adopter trading: the lagged-return component and the exogenous news shock. The decomposition matters because the rolling AR rule of section 3.3 is, by construction, an estimate of the AR component of the return process and therefore of x_{t+1} in expectation, while the only return a deployed forecaster can observe and score against is r_{t+1}. Both targets are therefore primary endpoints with asymmetric interpretations. Realised-return R^2 scores the forecast against r_{t+1} of (12): the total performance a deployed forecaster actually sees, including whatever part of the return coordinated adopter orders have pushed in the forecast direction through μD_t, the reading expected to rise with adoption (the self-fulfilling channel). Demand-adjusted R^2 scores the same forecast against x_{t+1} of (13): the counterfactual that strips out the contemporaneous price impact and asks whether the independent signal the forecast was designed to exploit survives once self-fulfilment is removed, the reading expected to fall (the demand-adjusted-erosion channel).

One definitional caveat is worth stating exactly. The demand-adjusted return is *not* the full regression residual of an AR(1) fit on returns: under the data-generating process (3), that residual is the pure exogenous news shock, $\sigma\,\varepsilon_{t+1} = r_{t+1} - \varphi\, r_t - \mu\, D_t$. The quantity x_{t+1} removes the contemporaneous adopter-demand term μD_t only, leaving the lagged-return persistence channel φr_t in place. It is best read as a counterfactual: the return path the market would have produced had adopter demand not moved the quote in the current period, so that the predictable AR(1) component still has a chance to show up in the next-period observation. The distinction names the erosion channel: against $\sigma\varepsilon_{t+1}$, independent of everything known at time t, any forecast carries zero signal in expectation whatever adoption does, while x_{t+1} retains exactly the predictable component the forecast was built to recover, so a fall in demand-adjusted R^2 records the erosion of an independent signal that was genuinely there. That is the claim "demand-adjusted erosion" makes, and the reading tracked from section 5.3 onward.

Table 1: Recurring notation, gathered here as a recap of the model. Each symbol is defined where it first appears alongside the equations of sections 3 and 4.

Symbol	Meaning
r_{t+1}	realised one-period return of the log-price, $r_{t+1} = p_{t+1} - p_t$
D_t	aggregate signed demand, the per-trader average order
μ	price-impact coefficient of the market maker
φ	input AR(1) coefficient of the return law
π	per-period entry probability under stochastic diffusion
A_t	adoption share, the fraction of traders using the shared rule
A^*	critical adoption share, one threshold per evaluation target
x_{t+1}	demand-adjusted return, $x_{t+1} = r_{t+1} - \mu D_t$
w	rolling estimation window length of the forecasting rule
p	AR order of the forecasting rule (subscripted p_t is the log-price)

4. Metrics and experimental design

4.1 Primary forecast-performance endpoints

4.1.1 Realised-return R^2

The first primary endpoint is the standard out-of-sample R^2 of the one-step forecast (7) against the realised return, the target a deployed forecaster can observe:

$$R^2_{\text{realised}}(t) \;=\; 1 \;-\; \frac{\sum_{s=t-W}^{t-1}\left(r_{s+1}-\hat{r}^{(A)}_{s+1}\right)^2}{\sum_{s=t-W}^{t-1}\left(r_{s+1}-\bar{r}\right)^2}, \tag{14}$$

where the sums run over the most recent W forecast-target pairs ending at period t and $\bar{r}$ is the mean realised return over that window. The evaluation window is distinct from the AR estimation window w; every reported run uses $W = 1000$ periods, the notebooks' `eval_window` parameter.

The forecasts entering (14) are strictly out of sample, each $\hat{r}^{(A)}_{s+1}$ fitted on returns through period s only. One convention is worth stating once: the out-of-sample R^2 benchmarks against the trailing window's in-sample mean of the target, not a real-time prevailing mean. The choice slightly favours the benchmark, and with near-zero-mean returns the difference is negligible. Because the target contains the contemporaneous impact term μD_s of (12), this is the total-performance reading of section 3.6, expected to rise with adoption through the self-fulfilling channel.

4.1.2 Demand-adjusted-return R^2

The second primary endpoint scores the same forecast, on the same path and trailing window, against the demand-adjusted return x_{s+1} of (13):

$$R^2_{\mathrm{da}}(t) \;=\; 1 \;-\; \frac{\sum_{s=t-W}^{t-1}\left(x_{s+1}-\hat{r}^{(A)}_{s+1}\right)^2}{\sum_{s=t-W}^{t-1}\left(x_{s+1}-\bar{x}\right)^2}, \tag{15}$$

with $\bar{x}$ the window mean of the demand-adjusted return; the benchmark convention of section 4.1.1 applies unchanged. The same forecast can earn two different R^2 readings on the same path because the targets differ by exactly that impact term: adopter orders are aligned with the forecast, so μD_s systematically moves the realised target toward the forecast, while a rolling fit that has absorbed the resulting amplified autocorrelation over-predicts the demand-adjusted target. The two endpoints are reported on equal footing, neither being canonical: R^2_{realised} measures what a deployed forecaster sees, and R^2_{da} whether the independent signal the forecast was built to exploit survives once self-fulfilment is stripped out, the erosion channel whose thresholds section 4.3 defines.

4.2 Effective AR coefficient as a market-dynamics diagnostic

The two R^2 endpoints score the forecast; the rolling effective AR coefficient $\hat{\varphi}^{\mathrm{eff}}(t)$ instead tracks the dynamics of the market the forecast trades in. This single quantity recurs throughout the report under a few interchangeable names: *effective AR coefficient*, *effective coefficient*, and *(measured) return persistence* all refer to $\hat{\varphi}^{\mathrm{eff}}(t)$, the rolling lag-1 autocorrelation of realised returns. Estimated from realised returns alone, with no reference to the forecast, it is computed at each period as the lag-1 sample autocorrelation of the most recent W realised returns, with the same trailing $W = 1000$ as the R^2 endpoints in every reported run. With no adoption it should sit at the input φ of the return law (3); section 5.1 verifies that it does, with no drift. Under adoption it should rise above the input value: adopter demand responds positively to recent returns through the shared forecast, so the impact term μD_t reinforces the lagged-return component and measured persistence grows, the mechanism behind the realised-R^2 climb of section 5.3.

It is a diagnostic rather than a third endpoint because its role is explanatory: the over-prediction mechanism of the previous subsection runs through this absorbed amplification, so the rise of $\hat{\varphi}^{\mathrm{eff}}$ is the mechanical reason the two R^2 readings diverge. Sections 5.3 and 5.4 track it against the adoption share, and the relative threshold $A^*_{\varphi,\mathrm{rel}}$ of section 4.3 summarises its rise across the parameter grid of section 5.5.

4.3 Target-specific critical adoption shares

A summary of the form "the rule stops working at adoption share A^*" is ambiguous, because the two R^2 endpoints of the preceding subsections are expected to move in opposite directions on the same path. The design therefore defines one critical adoption share per evaluation target; Table 2 collects the family. Section 5.5 reports where each threshold fires across the parameter grid, turning the dual-channel claim into a quantitative statement about which targets erode, at what share, and where in parameter space.

Table 2: The threshold family. A is the adoption share; each threshold is the smallest A at which the stated criterion is met. Relative thresholds (subscript rel) compare a rolling metric to that metric's own mean over the run's low-adoption baseline window; their factors were fixed before the phase 6 sweep was run.

Symbol	Target	Definition
$A^*_{R^2,\text{realised}}$	realised-return R^2 (14)	Primary (absolute): rolling realised-return R^2 crosses zero from above. Typically not reached, because realised R^2 rises with adoption; the absence is itself the realised-channel finding.
$A^*_{R^2,\text{realised,rel}}$	realised-return R^2	Operational analogue: realised R^2 falls to half its low-adoption baseline. Fires only as noise crossings in cells whose baseline is near zero.
$A^*_{R^2,\text{da}}$	demand-adjusted R^2 (15)	Primary (absolute): rolling demand-adjusted R^2 falls through zero.
$A^*_{R^2,\text{da,rel}}$	demand-adjusted R^2	Operational analogue: demand-adjusted R^2 falls to half its low-adoption baseline. The headline erosion threshold of section 5.5.
$A^*_{\varphi,\text{rel}}$	effective AR coefficient (§4.2)	Relative only: rolling $\hat{\varphi}^{\text{eff}}$ rises to 1.5 times its low-adoption baseline; a growth criterion, read from below, on the market-dynamics diagnostic.

The relative qualifier marks an operational analogue, not a redefinition. The primary definitions are absolute zero crossings, but on the two forecast-performance targets the cost-free baseline gives them little to detect: realised R^2 rises with adoption, so $A^*_{R^2,\text{realised}}$ goes unreached, and demand-adjusted R^2 falls through zero only where price impact is strong or the baseline signal is itself near zero, while at moderate settings it falls yet stays positive. The relative thresholds make erosion detectable in that regime and were fixed before the sweep was run: a fall to half the metric's own low-adoption baseline on the two R^2 targets, a rise to 1.5 times baseline on the effective coefficient, each defined only where that baseline is positive and flagged by the rel subscript so it is never mistaken for a zero crossing.

4.4 Simulation protocol, seed handling, and parameter grid

Every experiment in section 5 is a set of end-to-end simulator runs with all parameters fixed in code in the owning notebook. Table 4 in Appendix D collects the headline parameters: the default column is the configuration shared by the adoption phases (4 to 7, with the exceptions its caption lists), the range column the values crossed by the phase 6 sweep or, for the adoption rate, by the phase 4 regime comparison.

Seed handling has three levels. Single-path runs fix one master seed: 42 for phase 1, 43 for phase 3, 71 for phases 4 and 5. Regimes meant to be compared share that seed: phases 4 and 5 re-instantiate the generator from seed 71 for each adoption regime, and the dedicated adoption generator of section 3.4 keeps the market shock stream identical across them, so cross-regime differences are attributable to adopter behaviour (paired shocks). Monte Carlo exercises derive per-run seeds deterministically from a notebook-level base seed: 100 seeds from base 0 in phase 2, 100 from 1000 and 50 from 5000 for phase 4's band and saturation figures, 10 per grid cell from 2000 in phase 6. The pairing extends into the sweeps. Phase 6's per-run seed is offset by the window index and the (φ, μ) cell but deliberately not by the AR order, so all four orders p run on identical shock streams.

The phase 6 sweep crosses five price-impact values, six input AR coefficients, three windows, and four AR orders: 360 cells at 10 seeds per cell, 3600 simulator runs. The sweep's window set differs from the design set of section 3.3: it drops the shortest window $w = 50$ and adds $w = 500$, bracketing the default $w = 250$ from both sides. Each run uses slow stochastic diffusion ($\pi = 3 \times 10^{-4}$), released only after a 1500-period low-adoption baseline window has completed, so the threshold searches of section 4.3 start from $A = 0$ against a baseline measured at zero adoption.

5. Results

5.1 Baseline market and benchmark validation

We first check that the simulator reproduces the baseline market of section 3.1 with every trader on the null rule, in a single run with $N = 200$, $T = 5000$, $\mu = 0.05$, input $\varphi = 0.10$, news scale $\sigma = 0.01$, null-order scale $\sigma_q = 1.0$, and seed 42. Null orders are independent Gaussian draws, so aggregate demand feeds the return only as unpredictable price-impact noise, leaving φr_t as the only exploitable structure. The run behaves accordingly (Figure 1): the log price wanders without systematic drift (mean one-step return 2.0×10^{-4}, within 1.4 standard errors of zero) and the return series is roughly stationary with standard deviation 0.0105. The empirical lag-1 autocorrelation of returns is 0.103, matching the input $\varphi = 0.10$ within the sampling error of roughly $1/\sqrt{T} \approx 0.014$.

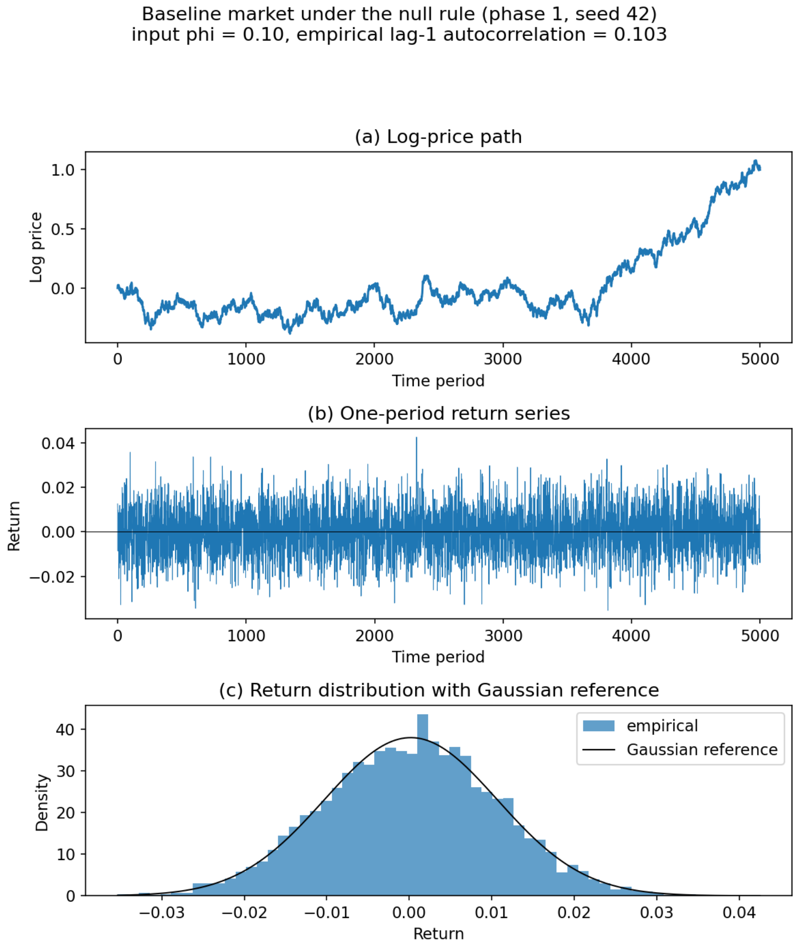

Figure 1: **The null-rule market is drift-free, near-Gaussian, and recovers the input autocorrelation.** Baseline market under the null rule (phase 1: $N = 200$, $T = 5000$, $\mu = 0.05$, input $\varphi = 0.10$, seed 42). (a) The log-price path wanders with no systematic drift. (b) The one-period return series is roughly stationary around zero. (c) The return histogram is symmetric and close to the Gaussian reference density built from the empirical mean and standard deviation. The empirical lag-1 autocorrelation of returns is 0.103, against the input 0.10.

Next, we rerun the null-rule market across 100 independent seeds (0 through 99) with $N = 128$, $T = 5000$, and the same market parameters, confirming the baseline is a stable control rather than a one-seed accident. The per-seed descriptive statistics cluster tightly around their theoretical values: the mean return averages 1.5×10^{-5} with a cross-seed standard deviation of 1.6×10^{-4}, consistent with a zero unconditional mean; the return standard deviation averages 0.0110 with spread 0.00012; the lag-1 autocorrelation averages 0.103 with spread 0.012, around the input $\varphi = 0.10$; and the excess kurtosis averages -0.0065 with spread 0.071, indistinguishable from the Gaussian value of zero. The kurtosis reading and the symmetric single-seed histogram from phase 1 support treating null-rule returns as approximately Gaussian, as expected with Gaussian news and null orders and demand averaged across many traders. The sharpest control statistic is the rolling effective AR coefficient (Figure 6, Appendix C): estimated on a 1000-period sliding window and averaged across seeds, the trace sits near the input value and stays flat, total range

0.016 over the 4001 periods with a complete window. This control underpins everything that follows: attributing later movements in forecast R^2 and effective AR coefficient to adoption is meaningful only because the same market without adoption shows no comparable drift over the same horizon.

5.2 AR forecast performance without adoption

We now attach the rolling AR(1) rule of section 3.3 to the null market, in offline mode: the forecast is estimated and scored, but no trader acts on it. A single run uses $N = 200$, $T = 8000$, $\mu = 0.05$, $\sigma = 0.01$, $\sigma_q = 1.0$, and seed 43, with the input coefficient raised to $\varphi = 0.25$ so the rule has a clearly detectable signal to recover. Each period the rule fits an AR(1) on the most recent $w \in \{50, 100, 250\}$ returns and issues a strictly out-of-sample one-step forecast, scored post hoc against realised returns by the rolling out-of-sample R^2 (14) and by full-sample mean squared forecast error (MSFE). With zero adoption the realised and demand-adjusted returns differ only by unpredictable null-order impact, so this subsection reports realised-return performance only; the targets separate once adopters start trading in phase 4.

The rule recovers the signal it was built for: the mean rolling out-of-sample R^2 is small, positive for every window, and rising with window length, 0.013 for $w = 50$, 0.033 for $w = 100$, and 0.047 for $w = 250$. In the time profile (Figure 7, Appendix C) the traces co-move because they score one path, and longer windows sit almost everywhere higher: the $w = 250$ trace is positive at every evaluated period (minimum 0.020), $w = 100$ in 98.5 percent of periods, while $w = 50$ dips below zero in roughly 16 percent. These levels are consistent with the return law's ceiling: an oracle knowing φ exactly would attain a population R^2 of $\varphi^2 = 0.0625$, which the longer windows approach from below as estimation noise shrinks. The coefficient estimates tell the same story (Figure 8, Appendix C): the rolling $\hat{\varphi}$ fluctuates around the input value, with means 0.195, 0.217, 0.229 and standard deviations 0.145, 0.103, 0.063 for $w = 50$, 100, 250; the mild downward gap reflects small-sample least-squares AR(1) bias plus single-path sampling noise, and closes as the window lengthens. Full-sample MSFE is nearly flat across windows (1.18×10^{-4} to 1.15×10^{-4}) and the rolling MSFE traces show no trend, as expected when no trader acts on the forecast. Sign accuracy, a diagnostic rather than an endpoint, is 0.56 for the two shorter windows and 0.57 for $w = 250$.

This completes the pre-adoption baseline: a market that does not drift without forecast trading, and a rule that recovers the linear-persistence signal at a known, stable level due entirely to the φr_t term.

5.3 Dual-channel result under stochastic adoption

We now turn the offline forecast into a tradeable rule and let its use spread. Adopters map the shared rolling AR(1) forecast ($w = 250$) into positions through the mean-variance demand (4), clipped per (5); non-adopters keep the null rule. With combined risk scaling $a\sigma^2 = 0.001$ and per-trader cap $\bar{q} = 0.05$ the cap binds in nearly every period, so adopters are effectively fixed-size traders on the forecast's sign, with impact comparable in magnitude to the φr_t term rather than dominating it. Adoption follows the stochastic diffusion of (8) and (9) in three regimes, zero adoption ($\pi = 0$), slow diffusion ($\pi = 3 \times 10^{-4}$), and fast diffusion ($\pi = 10^{-3}$), each with exit probability $\delta = 0$, on paired shocks from the shared seed 71 so that cross-regime differences are attributable to adoption alone. The market is otherwise the phase 3 configuration ($N = 200$, $T = 8000$, $\mu = 0.05$, input $\varphi = 0.25$, $\sigma = 0.01$, $\sigma_q = 1.0$), with every rolling metric on a trailing 1000-period window. Adoption is held at zero until period 1250, the first period with a complete metric window; the headline comparison contrasts a low-adoption window (periods 1250 to 2750) with a high-adoption window (the final 1500 periods), by which point the slow regime's mean adoption share is 0.836 and the fast regime's 0.994.

Against realised returns, the forecast improves as adoption rises: in the high-adoption window the fast regime's rolling out-of-sample R^2 against r_{t+1} is 0.190, versus 0.070 for the zero-adoption control on the same shocks (0.065 in the control's own late window), with the slow regime in between at 0.165. The control is the right low-adoption anchor: the fast regime's own early window already averages an adoption share of 0.505, with realised R^2 lifted to 0.094. The mechanism is the decomposition $r_{t+1} = \mu D_t + x_{t+1}$ of (12): every adopter trades in the direction of the same public forecast, so coordinated demand pushes the realised return in the forecast direction, and the forecast scores partly against its own price impact. The top-left panel of Figure 2 shows this is not a one-seed accident: pooled across 100 Monte Carlo seeds, mean realised-return R^2 rises monotonically in adoption share.

The same forecast on the same path is simultaneously losing the signal it was built to exploit: against the demand-adjusted return $x_{t+1} = r_{t+1} - \mu D_t$ of section 3.6 (the counterfactual, not the full regression residual), the rolling out-of-sample R^2 falls to 0.039 in the fast regime's high-adoption window, versus 0.082 under the control, with the slow regime at 0.047. The independent predictive content that survives once self-fulfilment is removed thus weakens as adoption rises: in the top-right panel of Figure 2 the Monte Carlo mean declines, the drop steepening at high shares.

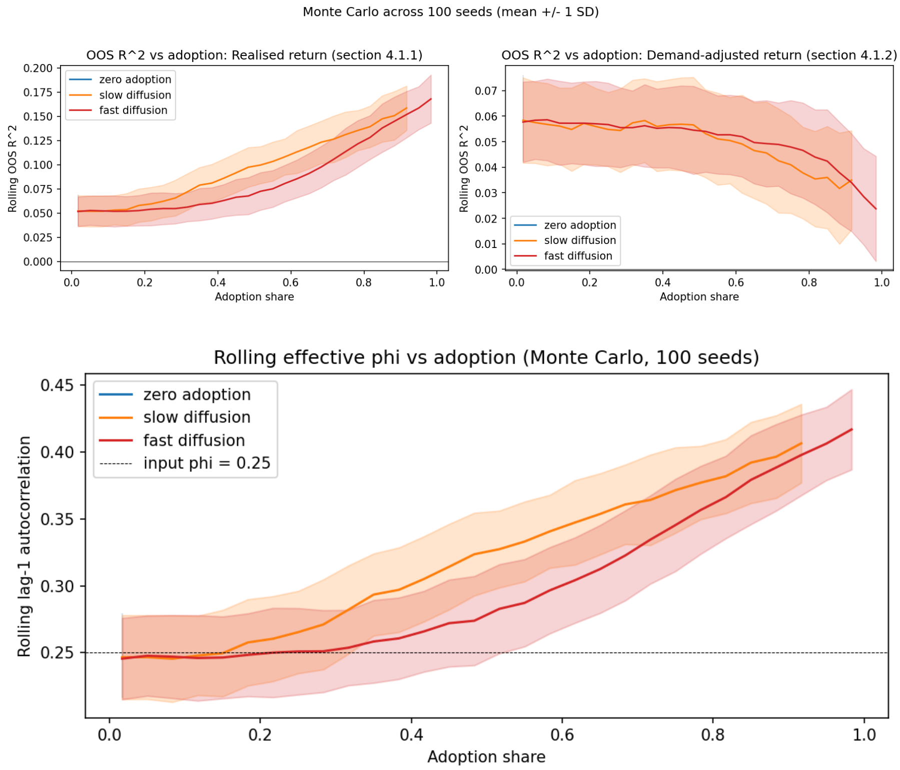

Figure 2: **Adoption raises realised R^2 while eroding demand-adjusted R^2, as measured persistence climbs.** Dual-channel Monte Carlo diagnostics versus adoption share, pooled across 100 seeds. Top: rolling out-of-sample R^2 of the shared AR(1) forecast against the realised return r_{t+1} (left, self-fulfilling channel) and against the demand-adjusted return $x_{t+1} = r_{t+1} - \mu D_t$ (right, demand-adjusted-erosion channel); note the different y-axis scales. Lines are means within 30 adoption-share bins pooling post-warm-up periods across seeds, bands ± 1 standard deviation; bins with 10 or fewer observations are dropped, trimming the slow regime's curves short of full adoption, and the zero-adoption regime occupies a single bin at share zero, so it appears in the legends but draws no visible curve. Mean realised R^2 rises from about 0.05 near zero adoption to about 0.17 at full adoption while mean demand-adjusted R^2 falls from about 0.06 to about 0.02; the slow-diffusion curves sit slightly left of the fast ones because the trailing 1000-period metrics lag the instantaneous share, more so the faster adoption ramps. Bottom: rolling lag-1 autocorrelation of realised returns versus adoption share, same design and binning; the dashed line marks the input $\varphi = 0.25$. Both diffusion regimes start close to the input near zero adoption and rise steadily, reaching roughly 0.40 to 0.42 at their highest observed shares.

The rolling effective AR coefficient of section 4.2 is the mechanical link between the two readings: it rises from 0.282 under the control, close to the input $\varphi = 0.25$, to 0.447 in the fast regime's high-adoption window (Figure 2, bottom panel). Adopter demand responds positively to recent returns through the forecast, so μD_t reinforces the lagged-return component and measured persistence grows with adoption; the rolling AR fit absorbs this amplified coefficient and systematically over-predicts the demand-adjusted process, whose own AR coefficient stays at the input 0.25 by construction. That over-prediction is the demand-adjusted decline of the previous paragraph. The control meanwhile moves far less across the same two windows (realised 0.070 to 0.065, demand-adjusted 0.082 to 0.068, effective coefficient 0.282 to 0.278), so the diffusion-regime movements are adoption effects, not time effects (single-path time profiles in Figure 9, Appendix C).

This is the dual-channel result, the headline finding of the report: adoption raises realised-return predictive performance through the self-fulfilment channel while eroding the independent demand-adjusted signal, on the same path and from the same forecast; which reading a forecaster sees depends entirely on whether the evaluation target includes the forecaster's own price impact. The separation is a steady state of the saturated market, not a transient of the adoption ramp: extending the fast-diffusion regime to $T = 30000$ periods across 50 Monte Carlo seeds, roughly 25000 of them at full adoption, the readings hold their separated levels throughout, the cross-seed realised mean near 0.17 and the demand-adjusted mean near 0.02, with no drift back together (Figure 3).

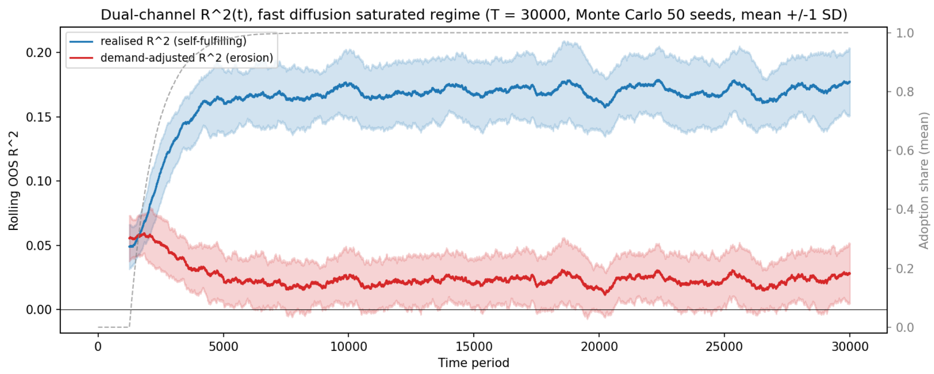

Figure 3: **The dual-channel separation is a steady state, not a transient of the adoption ramp.** Dual-channel R^2 over an extended fast-diffusion horizon ($T = 30000$, mean ± 1 standard deviation across 50 Monte Carlo seeds, trailing 1000-period metric window): rolling out-of-sample R^2 against the realised return (blue, self-fulfilling channel) and against the demand-adjusted return (red, erosion channel), with the mean adoption share (grey dashed, right axis) saturating early in the run. After the ramp the two readings hold their separated levels for the rest of the horizon, the realised mean near 0.17 and the demand-adjusted mean near 0.02: the dual channel is a steady state of the saturated market, not a transient of the transition.

5.4 Endogenous CE-based switching

We next make adoption respond to how the forecast is actually paying: stochastic diffusion is replaced by the certainty-equivalent switching of section 3.4, scored on the realised one-period profit $q_{i,t}\, r_{t+1}$ of one representative trader per rule, the common capped adopter position and a representative non-adopter's order. Each rule's certainty equivalent (10) is computed over $W_A = 500$ periods with $a = 1$, and each non-adopter switches to the advanced rule with the logistic probability (11) in the score gap $S_t = \mathrm{CE}_t^{(A)} - \mathrm{CE}_t^{(0)}$, with $\alpha = -5$ and $\beta = 10000$. At a zero score the implied switch probability is $\Lambda(-5) \approx 0.0067$ per period, and score movements of order 10^{-4} move it strongly in either direction. Everything else stays at the phase 4 configuration with adoption starting at period 1250; the comparator is the fast stochastic regime ($\pi = 10^{-3}$)

rerun on paired shocks from seed 71, so any difference is attributable to the adoption mechanism.

The adoption path changes shape; the erosion does not. The endogenous share is steplike: through most of the run-up the score hovers slightly below zero (negative in 88 percent of pre-saturation periods, held down in part by the null rule's positive own-impact profit mean $\mu\sigma_q^2/N = 2.5\times10^{-4}$), and at the median pre-saturation score the implied switch rate is 0.00049 per period, about half the comparator's diffusion rate, so the share mostly creeps; episodes of positive score raise the per-period inflow by a factor of about 3.6, producing the bursts. The endogenous run reaches full adoption by period 2842; the stochastic share crosses 0.99 only at period 6164 (Figure 10, Appendix C). Yet over the same low- and high-adoption windows as the previous subsection the two regimes give nearly identical erosion readings: the demand-adjusted out-of-sample R^2 falls from 0.071 to 0.039 under stochastic diffusion and from 0.076 to 0.037 under performance switching, while the rolling effective AR coefficient rises from 0.320 to 0.447 and from 0.315 to 0.449 respectively, at mean adoption shares of 0.505 to 0.994 and 0.455 to 1.000. Figure 4 traces both diagnostics against the adoption share itself: the two regimes start from the same pre-adoption values, decline and rise along overlapping paths, and converge at full adoption. The performance-switching curve sits somewhat above the stochastic one on the demand-adjusted panel at intermediate shares, a lag artefact: the endogenous run sweeps those shares too quickly for the trailing 1000-period metric window to refresh (235 periods between shares 0.6 and 0.95, against 1945 for the diffusion run).

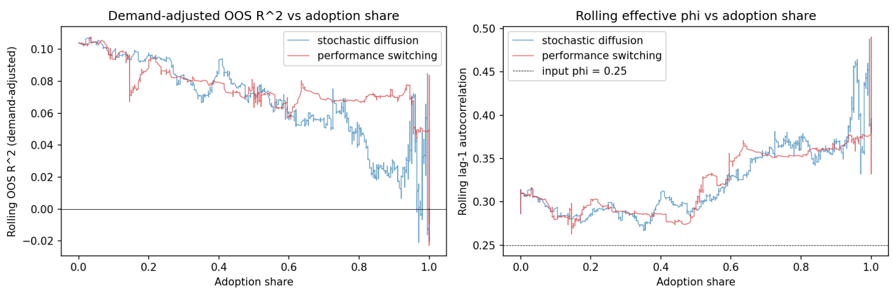

Figure 4: **Exogenous and endogenous adoption erode the signal along the same path.** Rolling out-of-sample R^2 against the demand-adjusted return (left) and rolling lag-1 autocorrelation of realised returns (right) versus adoption share, for exogenous stochastic diffusion ($\pi = 10^{-3}$) and endogenous CE-based switching on paired shocks (seed 71), trailing 1000-period windows. The regimes share the pre-adoption segment (demand-adjusted R^2 near 0.104, effective coefficient 0.286 to 0.314 at zero adoption, against the input $\varphi = 0.25$ marked by the dashed line), then decline (left) and rise (right) along overlapping paths, converging at full adoption. The switching curve sits above the stochastic one on the left at intermediate shares because the endogenous run sweeps those shares in few periods relative to the metric window; the vertical segments at share 1.0 collect each run's post-saturation stretch.

Endogenous switching therefore changes when the market visits each adoption level, not what adoption does once there: at matched shares both mechanisms produce the same demand-adjusted decline and effective-coefficient amplification, so the dual-channel result is not an artefact of the exogenous diffusion assumption; erosion is a property of the adoption share, however generated. The run also shows why the mechanism reinforces rather than brakes adoption in the cost-free baseline: adopter trades strengthen the autoregressive component of realised returns, so the advanced rule's certainty equivalent pulls level and then ahead as adoption builds (the score's pre-saturation maximum arrives one period before full adoption), and nothing pushes the score decisively negative at high adoption (Figure 11, Appendix C).

5.5 Parameter sweep and critical adoption shares

Finally, we turn the single-configuration findings above into a map. The sweep crosses $\mu \in \{0.025, 0.05, 0.075, 0.10, 0.15\}$, $\varphi \in \{0.05, \ldots, 0.30\}$, windows $w \in \{100, 250, 500\}$, and AR orders $p \in \{1, 2, 5, 10\}$, AR(1) the baseline and higher orders a robustness extension, at 10 seeds per cell (3600 runs). Everything else follows the phase 4 design ($N = 200$, $T = 8000$, trailing 1000-period metrics) under slow stochastic diffusion ($\pi = 3 \times 10^{-4}$, $\delta = 0$); adoption stays at zero until a 1500-period baseline window has completed, so each run measures its own pre-adoption baseline, and by run end the share reaches roughly 0.8. Seeds are paired across p (section 4.4), so differences across p isolate the forecast specification.

The zero-crossing threshold $A^*_{R^2,\mathrm{da}}$ fires only where price impact is strong or the baseline signal is near zero; at the settings used above the demand-adjusted R^2 falls but stays positive. We therefore report the relative analogues of section 4.3, fixed before the sweep was run and defined wherever the low-adoption baseline is positive: $A^*_{R^2,\mathrm{da,rel}}$ and $A^*_{R^2,\mathrm{realised,rel}}$ at half the metric's own low-adoption baseline (threshold factor 0.5), and $A^*_{\varphi,\mathrm{rel}}$ where the rising effective coefficient reaches 1.5 times its baseline, searched past the baseline window. A cell reports the mean crossing share over its crossing seeds, and only when at least half of the 10 seeds cross, so a noisy minority cannot set it. One caveat applies throughout: the rolling metrics trail adoption by up to the 1000-period evaluation window, so each A^* is a detection point, an upper bound on the erosion onset, not the onset itself.

The top panel of Figure 5 maps the demand-adjusted threshold for the AR(1) rule: it is reached in 67 of the 90 cells, at a mean detected share of 0.29. Two regularities organise the map. The crossing arrives later as the input φ grows, from very low shares in the $\varphi = 0.10$ row to 0.37 to 0.65 at $\varphi = 0.30$: the threshold scales with a baseline that grows with φ, so stronger signals need more adoption to lose half their content. And the crossing is roughly flat in μ (column means 0.25 to 0.31, no monotone trend), because the baseline R^2 also grows with μ and the relative threshold scales with it. The μ gradient lives instead in the depth of the erosion: the mean demand-adjusted R^2 over the final 1500 periods falls with μ in every φ row (Figure 13, Appendix C), from $+0.08$ at $(\varphi = 0.30, \mu = 0.025)$ to -0.28 at $(\varphi = 0.05, \mu = 0.15)$ for $w = 250$, so at the strongest price impact the independent signal is not merely halved but pushed below zero, the absolute criterion of section 4.3. The effective-coefficient threshold (bottom panel) is reached in 82 of 90 cells at a mean share of 0.38, and there the μ gradient shows in the threshold itself: at moderate-to-high φ, larger μ pulls $A^*_{\varphi,\mathrm{rel}}$ earlier ($w = 250$, $\varphi = 0.25$: 0.66 at $\mu = 0.05$, 0.33 at $\mu = 0.15$). The eight unreached cells sit at low μ and $\varphi \geq 0.20$, where feedback is too weak to amplify an already strong coefficient by half.

The realised-return threshold is the designed non-result: $A^*_{R^2,\mathrm{realised,rel}}$ is reached in only 28 of the 90 AR(1) cells, always early (mean share 0.06, maximum 0.21), with 26 of the 28 hits at $\varphi \leq 0.20$ and a median baseline realised R^2 of 0.014 at the hit cells. These are noise crossings, not erosion: where the baseline realised R^2 is near zero, the half-baseline threshold sits at essentially zero and rolling noise can dip below it without any systematic decline. The substantive realised-channel finding is the absence of crossings elsewhere: the tail-window realised R^2 rises with both φ and μ, up to 0.32 at the strongest cell (Figure 12, Appendix C), the self-fulfilment channel of phase 4 at grid scale. The same caveat covers the bottom rows of the demand-adjusted map: at $\varphi \leq 0.10$ the demand-adjusted baseline is also near zero, and those rows fail the filter or fire at near-zero shares.

The dual-channel pattern survives every AR order tested. At each p the effective-coefficient threshold is reached on 80 to 82 of 90 cells at mean shares 0.38 to 0.41, the tail demand-adjusted R^2 deepens with μ, and the realised threshold stays sparse (15 to 28 cells) and confined to low shares (Figures 14 and 15, Appendix C). Across the full four-order grid the demand-adjusted threshold is reached in 196 of 360 cells (54 percent, per-seed hit rate 0.53) at a mean share of 0.27, against a realised-channel hit rate of 0.26. Reachability falls with order (67, 59, 42, 28 of 90 cells for $p = 1, 2, 5, 10$) because the baseline falls with it (grid means 0.035, 0.028, 0.004, -0.041): a higher-order rule fits more coefficients on the same window, starts with less out-of-sample signal, and more often fails the positive-baseline requirement or the hit-rate filter. Where both AR(1) and AR(p) set a value, the delta-from-AR(1) map (Figure 16, Appendix C) is mildly negative on average (-0.04 at $p = 2$, -0.14 at $p = 5$, -0.20 at $p = 10$): higher-order rules cross somewhat earlier, consistent with having less signal to lose, and nowhere does the mechanism reverse. The erosion of the independent signal is therefore a property of the mechanism, not of the AR(1) specification.

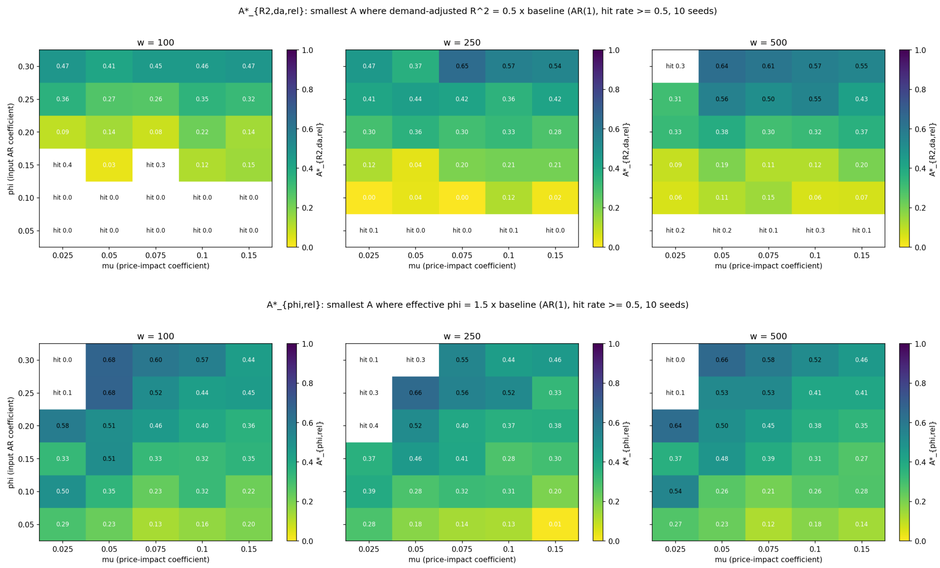

Figure 5: **The erosion threshold rises with input persistence; the persistence threshold falls as price impact grows.** Relative critical adoption shares across the (μ, φ) grid for the AR(1) rule, one panel per forecast window $w \in \{100, 250, 500\}$; cells where fewer than half of the 10 seeds crossed print their hit rate instead (for example "hit 0.3") and are left blank. Top: demand-adjusted threshold $A^*_{R^2,\mathrm{da,rel}}$ (rolling demand-adjusted R^2 at half its low-adoption baseline), averaged over the crossing seeds. The crossing share rises with the input φ, from very low values in the $\varphi = 0.10$ row, where the baseline is itself near zero, to 0.37 to 0.65 at $\varphi = 0.30$, and is roughly flat across μ; the $\varphi = 0.05$ row is unreached in every panel. Bottom: effective-coefficient threshold $A^*_{\varphi,\mathrm{rel}}$ (rolling lag-1 autocorrelation of realised returns at 1.5 times its low-adoption baseline), same layout and hit-rate convention. At moderate-to-high φ the threshold falls as μ grows (at $w = 250$, $\varphi = 0.25$: 0.66 at $\mu = 0.05$ down to 0.33 at $\mu = 0.15$); the unreached cells cluster at low μ and $\varphi \geq 0.20$, where price impact is too weak to amplify an already strong baseline coefficient by half, and the lowest φ row crosses earliest because a 50 percent relative increase on a small baseline is a small absolute move.

6. Discussion

6.1 Interpretation of the dual-channel mechanism

The dual-channel result of section 5.3 is one mechanism seen through two evaluation targets, not two findings that happen to coexist. Adoption makes the forecast self-fulfilling against the target a deployed forecaster observes: coordinated adopter demand moves the price in the forecast's direction, so the scored realised return becomes partly a product of the forecast itself (Figure 2, top-left panel). Against the demand-adjusted target, the counterfactual of (12) with contemporaneous price impact removed, the same adoption is purely degrading: the rolling fit calibrates to the amplified persistence of realised returns (bottom panel) and systematically over-predicts a process whose persistence has not changed. Self-fulfilment and erosion are two faces of the same coordinated order flow, so they arrive together on the same paths and persist across the sweep of section 5.5.

The practical lesson for a deployed forecaster is that realised-return R^2 over-reports the forecasting signal once the forecast has users with price impact. The realised reading conflates the stable signal the rule was trained to recover, the φr_t persistence of (3), with the price-impact channel created by its own adopters; as adoption grows, the headline number improves exactly while the independent content it appears to certify erodes. The top panel of Figure 5 maps the shares at which that content has lost half of its own low-adoption baseline. Realised evaluation therefore cannot distinguish a forecast that predicts the market from one the market has come to obey; the demand-adjusted audit that separates them requires exactly what an external evaluator usually lacks, the period's aggregate order flow D_t and the impact coefficient μ.

For theory, the result bears on the performative-prediction view of deployed models, in which measured accuracy depends on the behaviour the prediction induces and is commonly treated as splitting into a stable and a deployment-driven component. The simulator makes that split measurable in a market with price impact: the demand-adjusted target identifies the stable component, the gap between the two readings measures the deployment-driven one, and the threshold maps of section 5.5 chart the adoption needed to move each diagnostic by a fixed fraction of its baseline.

6.2 Why the effective AR coefficient rises with adoption

The rise in measured persistence in section 5.3, from 0.282 under the zero-adoption control to 0.447 in the fast regime's high-adoption window, follows mechanically from three chained ingredients of section 3. First, at $p = 1$ the shared forecast (7) is affine in the latest return, with an estimated slope tracking the fitting window's lag-1 autocorrelation, noisy but centred well above zero in these runs, so the forecast typically carries the sign of r_t when the lagged term outweighs the small intercept. Second, adopters turn the forecast into demand, though not in proportion to its magnitude: the cap in (5) binds in nearly every period at the phase 4 settings, so each adopter trades a fixed size $\bar{q}$ in the forecast's direction and the adopter block adds $A_t\bar{q}$, signed typically by r_t, to aggregate demand (1); sign-following rather than proportional demand changes the strength of what follows, not its direction. Third, the impact term μD_t enters the return law (3) with a positive sign, so the coordinated block pushes r_{t+1} in the direction of r_t.

A next-return component that co-moves with the current one is, by construction, extra lag-1 autocorrelation in realised returns, exactly what the diagnostic of section 4.2 computes. Its amplitude grows with the adopter mass, of order $\mu A_t\bar{q}$, so the measured coefficient climbs steadily with adoption rather than jumping (Figure 2, bottom panel). A naive reading predicts the opposite, that market-maker absorption of order flow should damp persistence; it fails because μD_t adds to the next-period return rather than offsetting it. Being structural, the rise appears under both adoption schemes at matched shares (section 5.4) and across the grid of section 5.5.

6.3 Boundary conditions and the realised-channel non-result

The realised-return channel shows no systematic erosion in any regime tested, an outcome the design anticipated with a dedicated threshold: $A^*_{R^2,\text{realised,rel}}$ is reached in only 28 of the 90 AR(1) cells, against 67 for the demand-adjusted threshold, with per-seed hit rates of 0.26 versus 0.53 over the full four-order grid (section 5.5). The hits sit at very low adoption, 23 of the 28 below $A = 0.10$, and in rows where the baseline realised R^2 is small: 26 of the 28 are at $\varphi \leq 0.20$, with a median baseline of 0.014 at the hit cells. They are the noise crossings of section 5.5: a near-zero baseline puts the half-baseline threshold at essentially zero, within reach of rolling noise.

The grid also bounds each positive claim: the demand-adjusted erosion, the core finding, holds in 67 of 90 AR(1) cells and 196 of 360 over all four AR orders, deepens with μ throughout, and is robust to adoption mechanism and AR order across paired-shock seeds (sections 5.4 and 5.5). The realised-channel amplification claim belongs to the high-μ, high-φ corner: at $\varphi \geq 0.25$ and $\mu \geq 0.10$ the tail-window realised R^2 reaches 0.19 to 0.32, exceeding its own baseline by 0.15 to 0.24. Outside that corner the amplification weakens: the tail realised R^2 remains clearly positive at the section 5.3 setting ($\varphi = 0.25$, $\mu = 0.05$) but flat, between -0.01 and 0.04, over the $\varphi \leq 0.10$, $\mu \leq 0.05$ block and the entire $\varphi = 0.05$ row at every μ tested: with no signal to amplify (baseline realised R^2 of -0.009), the channel never activates even at the strongest price impact. The lowest-φ rows bound the erosion claim from below: at $\varphi \leq 0.10$ the demand-adjusted baseline is near zero, leaving no signal to lose: the $\varphi = 0.05$ row never crosses and the $\varphi = 0.10$ row fires only at low shares.

6.4 Limitations

The findings carry two kinds of boundary: scope restrictions built into the minimal design, and implementation conventions that sections 6.2 and 6.3 lean on directly.

The scope restrictions are three. The market is single-asset, and linear price impact μD_t is the only channel from trading back into returns, so cross-asset and portfolio effects are out of scope and non-linear impact, such as a square-root law, is untested. The only forecasting rule is the rolling AR(p), with machine-learning and non-parametric forecasters excluded by design; time is discrete, so latency and other microstructure effects are not modelled. Non-adopters, finally, are a homogeneous pool of random null traders, with a heterogeneous ecology of trend followers and contrarians left as an unimplemented extension.

Two implementation conventions matter for the mechanism. The market maker absorbs position levels rather than position changes, the Beja and Goldman [1980] convention rather than Farmer and Joshi [2002] order flow, so a persistent position exerts persistent price pressure, strengthening the persistence mechanism of section 6.2. At the baseline risk scale and position cap, the clip in (5) binds in nearly every period, so adopters trade as fixed-size sign followers rather than graded mean-variance investors; relatedly, the switching rule's risk-aversion coefficient ($a = 1$) is set independently of the roughly 7 implied by the decision layer's risk scale at the realised return variance, keeping the certainty-equivalent penalty mild.

7. Conclusion

7.1 Summary of findings

The headline finding is the dual-channel result: adoption of the shared rolling AR(1) forecast lifts its realised-return out-of-sample R^2 from 0.07 under a zero-adoption control to 0.19 at near-full adoption, while on the same path its demand-adjusted R^2 falls from 0.08 to 0.04 and measured persistence rises from 0.28 to 0.45. One coordinated order flow drives both channels, and the pattern survives endogenous switching and a 3600-run sweep over price impact, input persistence, estimation window, and AR order. The realised-return channel itself never erodes systematically: its threshold fires in only 28 of 90 AR(1) cells, all weak-signal noise crossings at near-zero baselines.

7.2 Implications for forecasting under adoption

The practical lesson is about measurement. Realised-return R^2, the only score a deployed forecaster can compute, over-reports the forecast once it has users with price impact: it conflates the stable signal the rule was estimated to capture with the price impact of the rule's own adopters, and here it improves exactly while the independent content erodes. The demand-adjusted R^2 is the cleaner signal of whether the underlying forecasting hypothesis still holds, since it scores the forecast against a target with contemporaneous price impact removed; computing it requires the period's aggregate order flow and the impact coefficient, exactly what an external evaluator typically lacks. Where adoption is plausible, both readings should therefore be reported together: their gap measures the deployment-driven component of measured accuracy, the quantity the realised number silently absorbs.

7.3 Directions for future work

The most developed direction is the economic one. The natural next question is whether the forecast's statistical edge survives once trading is costly: a transaction-cost extension charging a proportional cost per unit traded and reading adopter net trading value against the no-skill null benchmark, the economic endpoint and its critical adoption share A^*_{profit} set out in the proposal. This extension was attempted during the project, the simulator records per-period profit and a preliminary cost sweep was run, but it was not carried far enough to report as a settled result: the null-relative comparison is confounded by the position-size asymmetry between the uncapped null rule and the capped adopter rule (the null books own-impact paper profit on a position roughly an order of magnitude larger than the adopter cap), which a faithful economic reading must first neutralise. A complete treatment, with a position-matched benchmark and an accounting that closes out positions rather than marking them at the post-impact quote, is left for future work.

Each remaining direction relaxes a limitation of section 6.4. Non-linear price impact, such as a square-root law, would test the mechanism against realistic impact concavity, and a heterogeneous ecology of trend followers and contrarians would set other strategies against the coordinated adopters. Machine-learning forecasters and multi-asset extensions would ask whether richer rules and cross-asset demand reproduce, dampen, or redirect the same two channels.

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A. Implementation

A.1 Software architecture

The simulator is a small pure-numpy Python package, `reflexive_market`, under `src/`: six working modules plus a seven-line `__init__.py` holding only a docstring and the version string, 588 lines in total. Four design constraints, fixed in the project working agreement before any simulator code was written, shaped the package. First, the mathematics uses numpy alone: AR estimation goes through `np.linalg.lstsq`, and no pandas, scipy, statsmodels, or scikit-learn import appears anywhere in `src/`. Second, every random draw flows through a `numpy.random.Generator` passed in as an explicit argument, never the global `np.random` state, so output is fully determined by the seed. Third, package functions never print or plot; parameters, figures, and the saved `npz` summaries live in the phase notebooks. Fourth, each module implements one small, named slice of the model, so every equation of section 3 traces to code; Table 3 gives the mapping.

Table 3: Modules of `reflexive_market`, their responsibilities, and the slice of the model each implements. Equation and section numbers refer to this report; line counts are from the final checkout.

Module	Lines	Responsibility	Reference
`market.py`	30	aggregate demand, market-maker price update, return law	eqs. (1)–(3)
`traders.py`	30	null random rule, demand mapping with position cap	eqs. (4), (5); §3.3
`forecast.py`	75	rolling AR(p) fit and out-of-sample forecast	eqs. (6), (7)
`adoption.py`	47	stochastic diffusion, certainty-equivalent switching	eqs. (8)–(11)
`metrics.py`	150	MSFE, out-of-sample R^2 (realised and demand-adjusted), effective AR coefficient	§4
`simulate.py`	249	`run()`: one function running T periods end to end	§3.5 timing

`simulate.py` is deliberately the largest module. Its single entry point `run()` is the only place where state evolves, and it follows the four-step intra-period timing fixed in section 3.5 exactly. Its defaults reproduce the phase 1 baseline with null traders only, so later phases extended the same function through keyword arguments (forecast window, risk scale and position cap, adoption parameters) rather than forking the loop; the two adoption mechanisms are mutually exclusive one-step pure functions invoked inside it. `metrics.py` scores completed runs after the fact; the target-specific critical adoption shares are computed in the phase notebooks instead because they need the full simulator output, not a return series alone.

A.2 Phased construction

The simulator was built in seven strictly ordered phases, a staging discipline fixed in the working agreement before any simulator code was written; phase N does not start until phase $N-1$ runs end to end, meets its recorded done-when condition, and is committed. The done-when criteria for phases 1 to 3 are unchanged since the scaffold; those for phases 4 to 7 were restated under the dual-channel framing after all seven phases had been implemented, once that framing had settled as the organising description of the result. The ordering enforces the design's central feasibility principle: the market must work before the forecasting rule is added, the rule before adoption, and extensions only once the baseline produces stable, interpretable results, so that any observed reflexivity effect traces to a small number of identifiable ingredients. Each phase owns one notebook (`notebooks/phase_NN_<name>.ipynb`) with every parameter in a single top cell, and saves its figures and `npz` summaries under `results/`. All seven phase notebooks exist; section 5 reports the results of phases 1 to 6 in build order, and the phase 7 transaction-cost extension is left for future work (section 7.3).

1. **Baseline market.** Implement the single-asset market with every trader on the random null rule and confirm the empirical lag-1 autocorrelation of returns matches the input φ (section 5.1).
2. **Benchmark validation.** Rerun the null market across many seeds to verify stable descriptive statistics and a drift-free rolling effective AR coefficient, the no-adoption control behind every later erosion claim (section 5.1).
3. **AR forecasting rule, offline.** Add rolling AR fitting and score the forecast against realised returns while no trader acts on it, recovering a small positive out-of-sample R^2 driven by the φr_t term (section 5.2).
4. **Forecast-based trading and stochastic adoption.** Let adopters trade the shared forecast while its use spreads by stochastic diffusion; the dual-channel result first appears here (section 5.3).
5. **Performance-based adoption.** Replace diffusion with certainty-equivalent switching and compare the two adoption mechanisms on paired shocks (section 5.4).
6. **Experiments and thresholds.** Sweep (μ, φ, w, p) and locate the relative critical adoption shares $A^*_{R^2,\mathrm{da,rel}}$ and $A^*_{\varphi,\mathrm{rel}}$, defined in section 4.3, across the grid (section 5.5).
7. **Evaluation and attempted extension.** Summarise the findings under the dual-channel framing; the transaction-cost extension and its economic endpoint were attempted but not fully conducted, and are left for future work (section 7.3).

The phases group into three blocks: phases 1 to 3 establish the pre-adoption control (a market with no forecast trading, a forecast with no market feedback, each verified on its own), phases

4 and 5 switch the feedback on and establish the dual-channel mechanism under two different adoption processes, and phase 6 quantifies the mechanism across the parameter grid. The phase 7 transaction-cost extension, intended to carry the analysis into economic terms, was attempted but not fully conducted and is left for future work (section 7.3).

A.3 Reproducibility

Every number and figure in this report can be regenerated from the repository checkout; three conventions carry the load. First, every notebook fixes its randomness in the parameters cell: single-run notebooks expose a scalar `seed` and multi-seed notebooks a `base_seed` and a seed count from which each per-run seed is derived deterministically, with the values listed in section 4.4. Because every draw flows through the explicitly passed generator (section A.1), the same seed and parameters reproduce the same run and figure; the test suite asserts this determinism directly. Second, every artifact names its origin: figures save to `results/figures/phase_NN_<name>.png` and numeric summaries to `results/data/phase_NN_<name>.npz`, and the `npz` files store the small arrays behind the report's tables and quoted numbers, so a reader can verify any of them without rerunning a simulation. Third, the environment is thin: the package requires Python 3.10 or newer and depends at runtime on numpy, matplotlib, and joblib alone, per the pure-numpy commitment of section A.1; `requirements.txt` adds only jupyter and pytest.

These conventions are enforced, not aspirational. The repository ships a 43-test pytest suite covering package import and per-phase invariants for phases 1 through 5, from seed determinism and recovery of the input φ to position-cap clipping and the mutual exclusion of the two adoption mechanisms. A GitHub Actions workflow installs the dependencies and the package on Python 3.11 and runs `pytest -q` on every push and pull request.

B. Model equations and derivations

This appendix collects the model's displayed equations in one place as a reading aid; each block re-displays equations from sections 3 and 4 under the report's own equation numbers, with symbols as in Table 1.

Market and return law, (1) to (3). Aggregate demand is the per-trader average order, the market maker moves the log-price linearly in it, and the return law adds a small autoregressive component:

$$D_t = \frac{1}{N}\sum_{i=1}^{N} q_{i,t}, \qquad p_{t+1} = p_t + \mu D_t + \sigma \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim \mathcal{N}(0,1),$$
$$r_{t+1} = \varphi r_t + \mu D_t + \sigma \varepsilon_{t+1}, \qquad |\varphi| < 1 \text{ small.}$$

Orders, (4) and (5). The mean-variance position and the capped adopter order it becomes are

$$z_{i,t} \;=\; \frac{\hat{r}_{i,t+1}}{a\,\sigma^2}, \qquad q^{(A)}_{i,t} \;=\; \mathrm{clip}\left(\frac{\hat{r}^{(A)}_{t+1}}{a\,\sigma^2},\, -\bar{q}_i,\, +\bar{q}_i\right);$$

a non-adopter submits the null order $q^{(0)}_{i,t} = \xi_{i,t} \sim \mathcal{N}(0,\sigma_q^2)$, uncapped, the final order is $q_{i,t} = (1-a_{i,t})\,q^{(0)}_{i,t} + a_{i,t}\,q^{(A)}_{i,t}$, and one-period profit is $\Pi_{i,t+1} = q_{i,t}\,r_{t+1}$ (sections 3.2 and 3.3).

Forecasting rule, (6) and (7). At time t the shared rule fits by least squares on the window $s = t-w,\ldots,t-1$ and issues a strictly out-of-sample one-step forecast:

$$r_{s+1} \;=\; c \;+\; \sum_{j=1}^{p} \varphi_j\, r_{s+1-j} \;+\; u_{s+1}, \qquad \hat{r}^{(A)}_{t+1}(w,p) \;=\; \hat{c}_{t,w,p} \;+\; \sum_{j=1}^{p} \hat{\varphi}_{j,t,w,p}\, r_{t+1-j}.$$

Adoption, (8) to (11). Stochastic diffusion enters and exits with fixed per-period probabilities; performance-based switching enters with a logistic probability in the certainty-equivalent gap:

$$\begin{aligned} P(a_{i,t+1}=1 \mid a_{i,t}=0) \;&=\; \pi, \qquad & P(a_{i,t+1}=0 \mid a_{i,t}=1) \;&=\; \delta, \\ \mathrm{CE}^{(k)}_t \;=\; \bar{\Pi}^{(k)}_t \;-\; \tfrac{a}{2}\,\hat{\sigma}^{2,(k)}_t, \qquad & & P(a_{i,t+1}=1 \mid a_{i,t}=0) \;&=\; \Lambda(\alpha + \beta\, S_t), \end{aligned}$$

with $S_t = \mathrm{CE}^{(A)}_t - \mathrm{CE}^{(0)}_t$ and $\Lambda(z) = 1/(1+e^{-z})$; the simulator runs exactly one mechanism at a time (section 3.4).

Evaluation decomposition, (12) and (13). The realised return splits into contemporaneous price impact and the demand-adjusted remainder:

$$r_{t+1} \;=\; \mu\, D_t \;+\; x_{t+1}, \qquad x_{t+1} \;=\; r_{t+1} \;-\; \mu\, D_t \;=\; \varphi\, r_t \;+\; \sigma\,\varepsilon_{t+1};$$

x_{t+1} removes the contemporaneous impact term only and is not the full regression residual $\sigma\varepsilon_{t+1} = r_{t+1} - \varphi r_t - \mu D_t$ (section 3.6). The forecast-performance endpoints score the same forecast against each target per (14) and (15).

Derived quantities used in the text. Under diffusion the adoption share moves at an expected per-period rate of $\pi(1-A_t) - \delta A_t$ (section 3.4). With no adoption, an oracle knowing φ exactly attains a population R^2 of φ^2 against realised returns, the ceiling of section 5.2.

C. Additional figures

This appendix collects the supporting figures referenced from sections 5.1 to 5.5, grouped into baseline and forecast diagnostics, adoption dynamics, and sweep robustness.

C.1 Baseline and forecast diagnostics

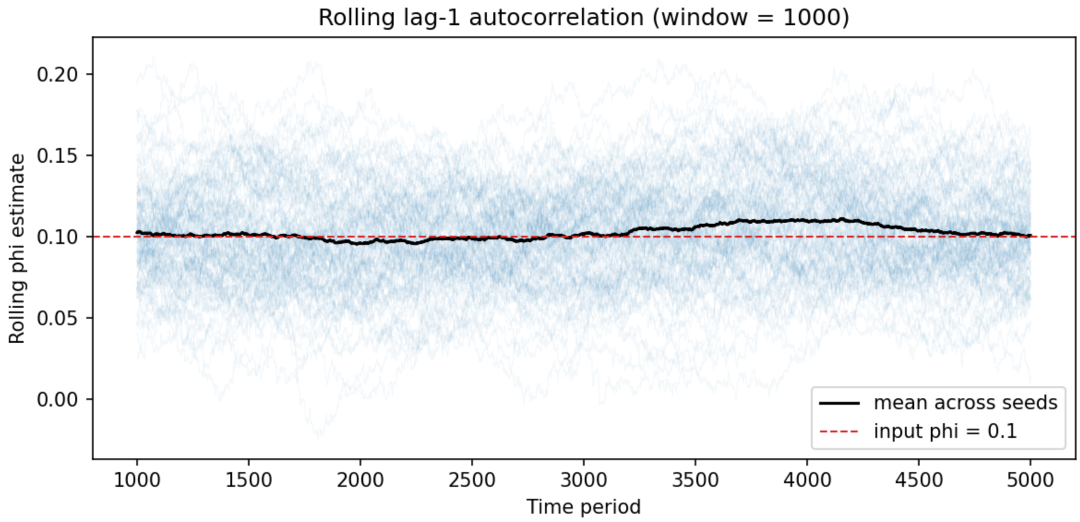

Figure 6: Rolling lag-1 autocorrelation of returns under the null rule, on a 1000-period sliding window for each of 100 seeds (faint lines); the black line is the cross-seed mean, the dashed line the input $\varphi = 0.10$. The mean trace stays flat for the full run, total range 0.016 over the 4001 periods with a complete window: no drift in the effective AR coefficient in the absence of adoption.

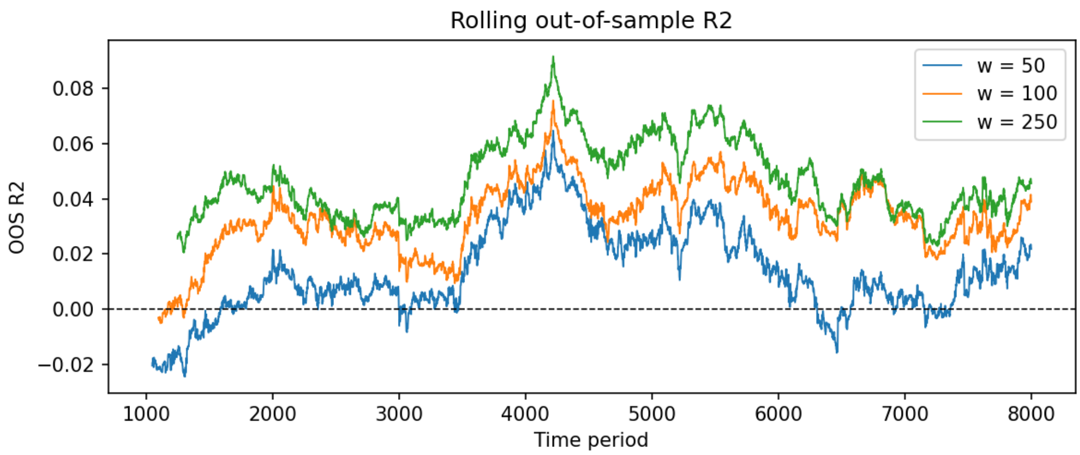

Figure 7: Rolling out-of-sample R^2 of the one-step AR(1) forecast against realised returns, on a trailing 1000-period window, for estimation windows $w = 50$, 100, and 250 on the same null-market path (input $\varphi = 0.25$, seed 43); the dashed line marks zero. The traces co-move because they score one path; longer windows sit almost everywhere higher ($w = 100$ and $w = 250$ briefly cross around periods 6800 to 6900), $w = 250$ stays positive at every evaluated period, and $w = 50$ intermittently dips below zero.

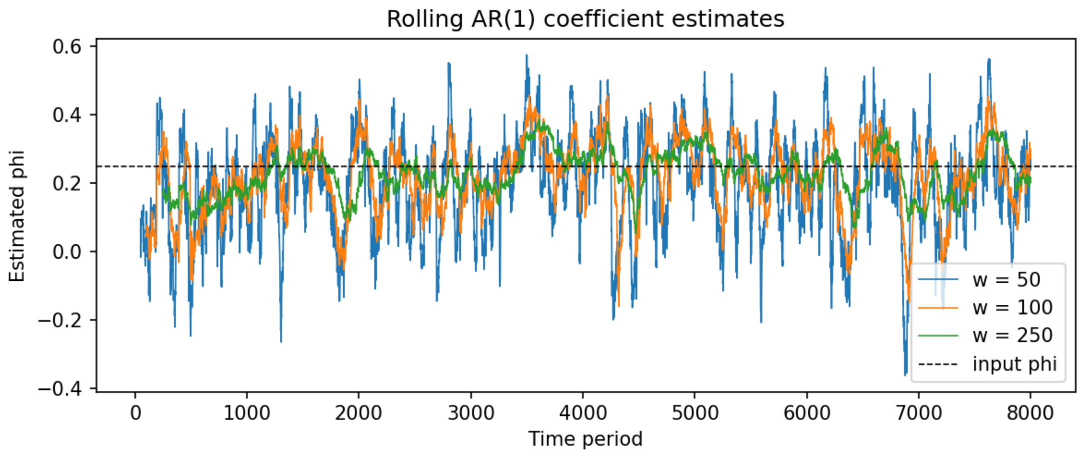

Figure 8: Rolling AR(1) coefficient estimates on the phase 3 null-market path (input $\varphi = 0.25$ dashed, seed 43) for estimation windows $w = 50$, 100, and 250. All three traces fluctuate around the input value with no trend, and shorter windows are visibly noisier (standard deviations 0.145, 0.103, 0.063 around means 0.195, 0.217, 0.229 for $w = 50$, 100, 250), the estimation noise behind the window ordering of Figure 7.

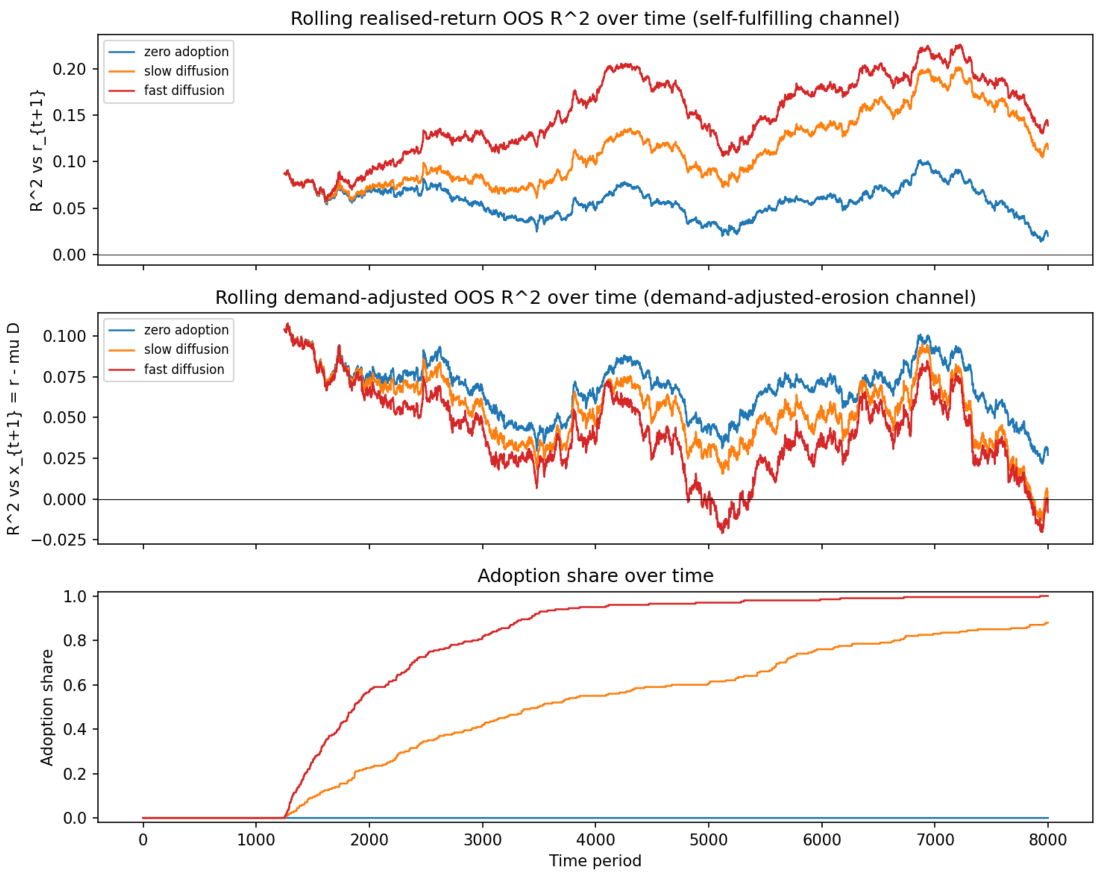

Figure 9: Single-path time profiles behind the phase 4 result (seed 71, paired shocks): rolling out-of-sample R^2 against the realised return (top) and against the demand-adjusted return (middle) for the zero-adoption, slow-diffusion, and fast-diffusion regimes, with each regime's adoption share below. After adoption is released at period 1250 the diffusion regimes' realised traces separate upward from the control while their demand-adjusted traces sink below it, the time-domain view of the bin-averaged Figure 2.

C.2 Adoption dynamics

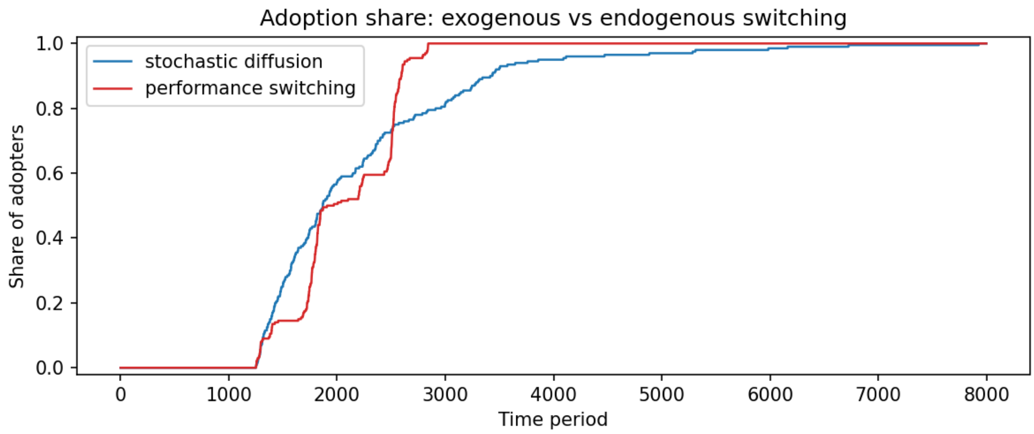

Figure 10: Adoption share over time for exogenous stochastic diffusion ($\pi = 10^{-3}$) and endogenous CE-based switching on paired shocks (seed 71). The endogenous share is steplike, creeping while the switching score is negative and bursting when it turns positive, and reaches full adoption by period 2842; the diffusion share ramps smoothly and crosses 0.99 only at period 6164.

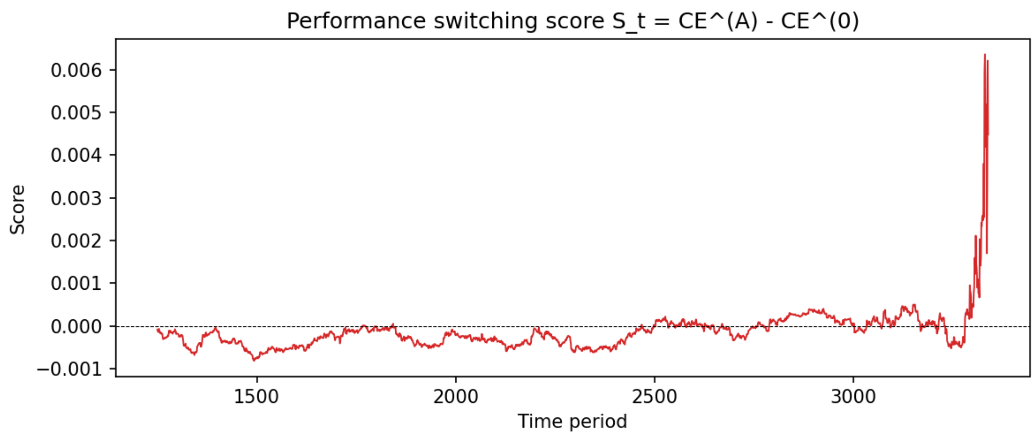

Figure 11: The switching score $S_t = \mathrm{CE}_t^{(A)} - \mathrm{CE}_t^{(0)}$ driving the endogenous run of Figure 10, shown from the adoption release at period 1250 until shortly after full adoption. The score hovers slightly below zero through most of the run-up, negative in 88 percent of pre-saturation periods, and the sharp spike to about $+0.0064$ arrives only after full adoption, once adopter trading has amplified the persistence the rule feeds on.

C.3 Sweep robustness

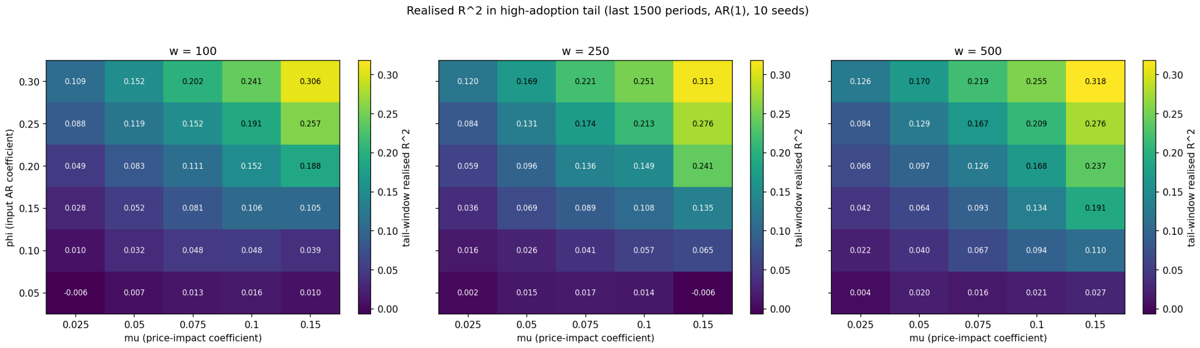

Figure 12: Mean realised-return R^2 over the high-adoption tail (final 1500 periods) across the (μ, φ) grid for the AR(1) rule, one panel per window w, 10 seeds per cell. The tail reading rises with both φ and μ in every panel, peaking at 0.31 to 0.32 in the top-right corner, while the $\varphi = 0.05$ row stays near zero: the self-fulfilment channel at grid scale and the realised-channel non-result of section 6.3.

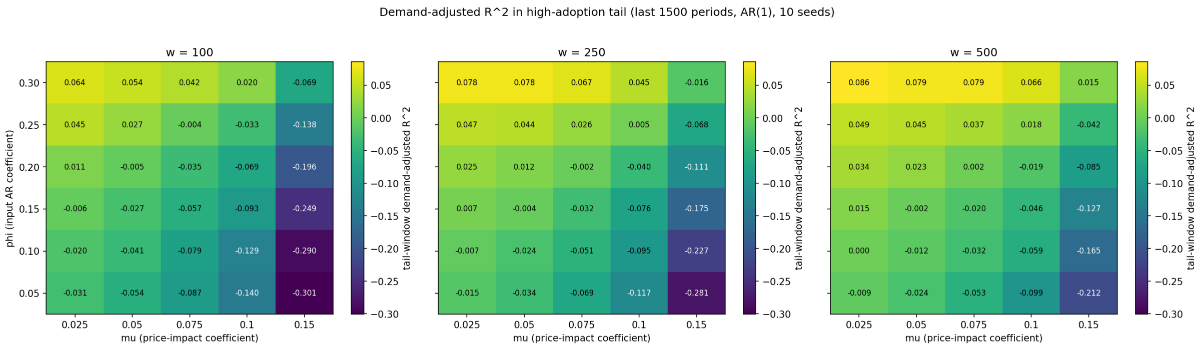

Figure 13: Mean demand-adjusted R^2 over the same high-adoption tail, same layout: the tail-depth view behind section 5.5. The reading falls with μ in every φ row, from positive values up to 0.06 to 0.09 at low μ and high φ down to -0.21 to -0.30 at $(\varphi = 0.05, \mu = 0.15)$ depending on the window: at the strongest price impact the independent signal is pushed below zero, not merely halved.

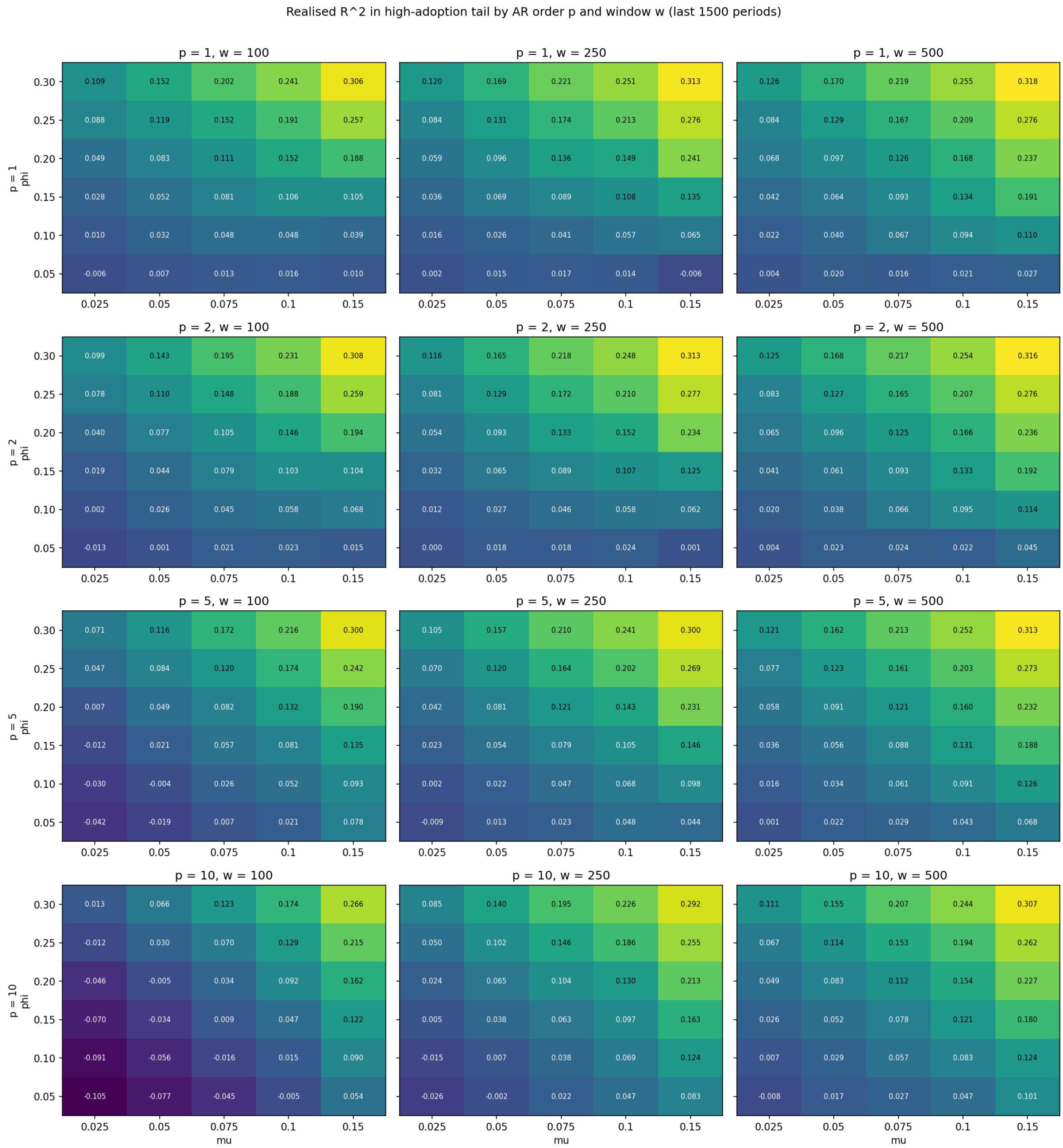

Figure 14: Tail-window realised-return R^2 by AR order p (rows) and window w (columns), twelve (μ, φ) panels. The rise in both φ and μ recurs at every order, every panel peaking at its high-φ, high-μ corner, with per-order maxima of 0.31 to 0.32, each at $w = 500$: self-fulfilment is not specific to the AR(1) rule.

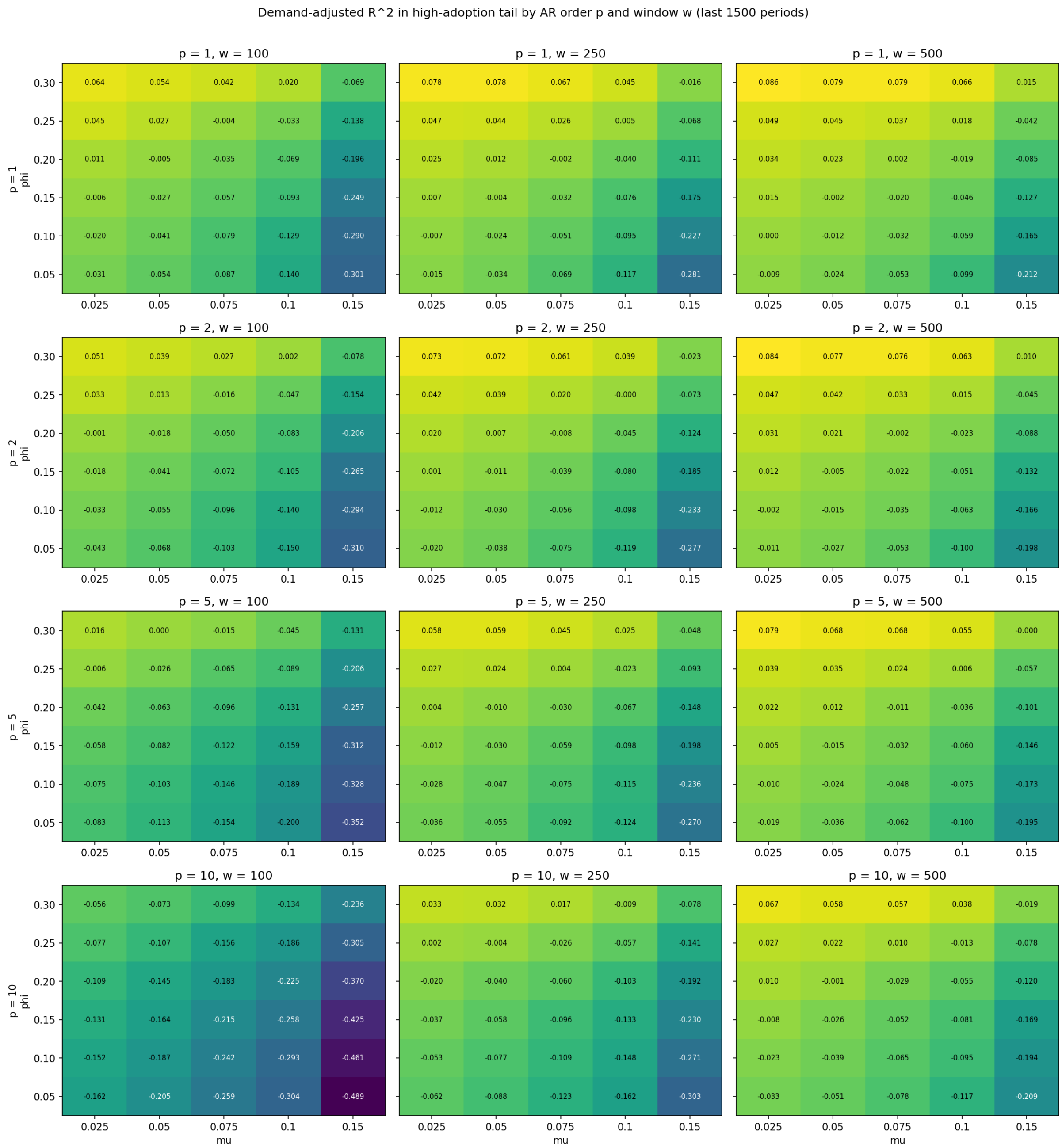

Figure 15: Tail-window demand-adjusted R^2 by AR order p and window w, same layout. Erosion deepens with μ in every panel and with the order overall, the deepest cell at -0.49 ($p = 10$, $w = 100$, $\varphi = 0.05$, $\mu = 0.15$), reflecting the lower out-of-sample baseline of higher-order rules noted in section 5.5.

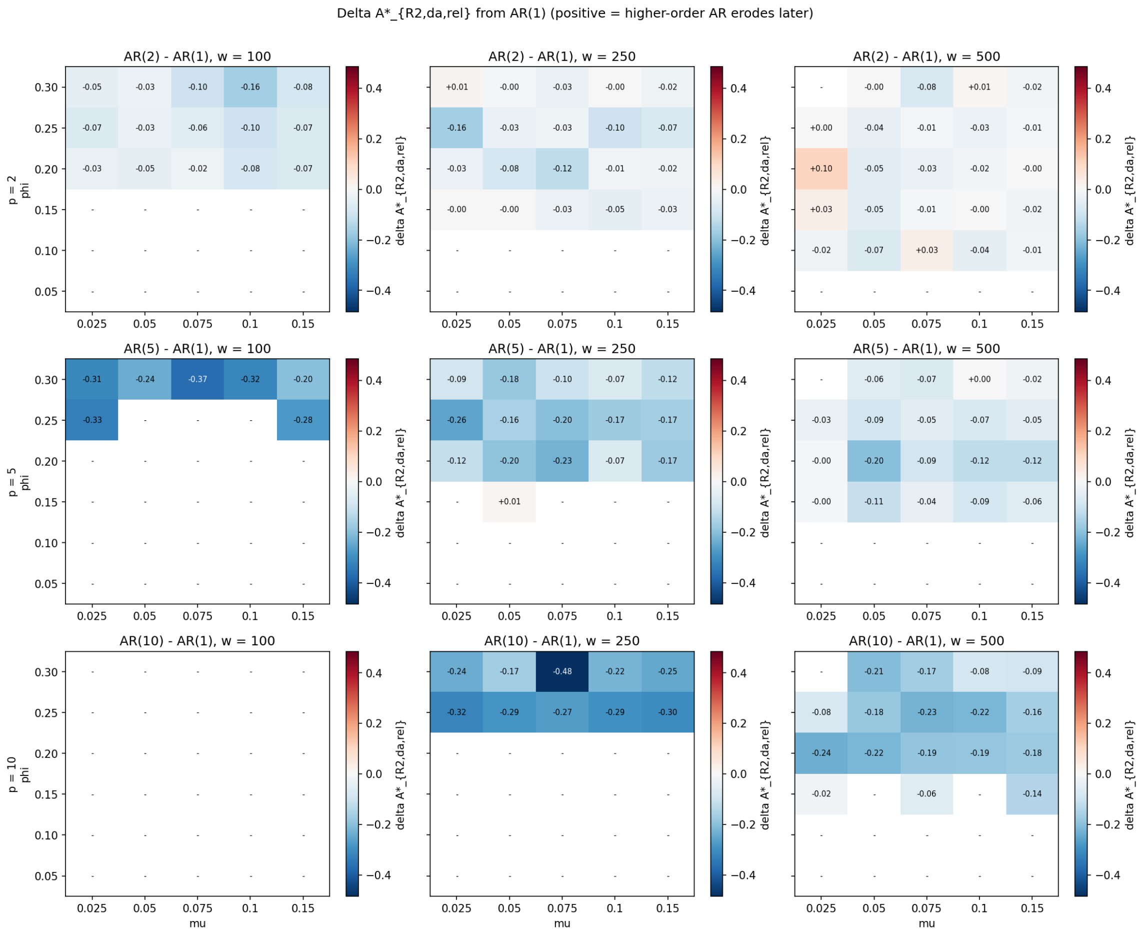

Figure 16: Cell-by-cell difference in the demand-adjusted threshold, $A^*_{R^2,\mathrm{da,rel}}$ at order p minus its AR(1) value, for $p \in \{2,5,10\}$ (rows) and $w \in \{100,250,500\}$ (columns); dashes mark cells where either side is undefined. The populated cells are predominantly mildly negative, with mean deltas of -0.04, -0.14, and -0.20 for $p = 2$, 5, 10: higher-order rules cross somewhat earlier, having less signal to lose, and nowhere does the mechanism reverse.

D. Parameter tables and reproducibility details

Table 4 collects the headline defaults and sweep ranges referenced from section 4.4; Table 5 expands them into the full per-phase configuration, one column per phase notebook.

Table 4: Headline simulation parameters. Defaults are the values shared by the adoption phases (4 to 7). The pre-adoption control phases differ in three places: phases 1 and 2 use $T = 5000$ and input $\varphi = 0.10$, phase 2 uses $N = 128$, and phase 3 fits windows $w \in \{50, 100, 250\}$ offline. Phase 4's saturation diagnostic extends fast-diffusion runs to $T = 30000$, and phase 5's stochastic comparator runs the fast rate $\pi = 10^{-3}$. The two shock scales are held fixed throughout, news at $\sigma = 0.01$ and null orders at $\sigma_q = 1.0$.

Parameter	Symbol	Default	Range tested
Trader count	N	200	held at default
Periods	T	8000	held at default
Price impact	μ	0.05	$\{0.025, 0.05, 0.075, 0.10, 0.15\}$
AR coefficient (input)	φ	0.25	$\{0.05, 0.10, 0.15, 0.20, 0.25, 0.30\}$
Forecast window	w	250	$\{100, 250, 500\}$
Forecast order	p	1	$\{1, 2, 5, 10\}$
Adoption rate (stochastic)	π	3×10^{-4}	$\{0, 3 \times 10^{-4}, 10^{-3}\}$
Risk scale	$a\sigma^2$	0.001	held at default
Position cap	$\bar{q}$	0.05	held at default

Reproducibility. The package requires Python 3.10 or newer, and continuous integration runs the 43-test pytest suite on Python 3.11 on every push and pull request (`.github/workflows/ci.yml`). All simulator mathematics is pure numpy: the runtime dependencies are numpy, matplotlib, and joblib alone, with `requirements.txt` adding only jupyter and pytest for the notebooks and tests. Every random draw flows through an explicitly passed `numpy.random.Generator` seeded by the conventions of section 4.4, so each run is fully determined by its notebook's parameter values. Each phase saves its figures as `results/figures/phase_NN_<name>.png` and its numeric summaries as `results/data/phase_NN_<name>.npz`; the npz files hold the arrays behind the report's tables and headline numbers, and the remaining quoted statistics are deterministic recomputations from the notebooks' fixed seeds, flagged as such where they appear.

Repository. The full source code (package `reflexive_market`, version 0.1.0), the seven phase notebooks, the test suite, and the saved figure and data artifacts all live in the project repository at `https://github.com/janwejchert/Capstone/tree/main`, laid out as described in appendix A.

Table 5: Per-phase simulation parameters, assembled from the parameters cell of each phase notebook. Dashes mark parameters a phase does not use; sweep entries refer to the phase 6 ranges of Table 4. Phase 5 runs two regimes on paired shocks, CE switching (no π) and the stochastic comparator at $\pi = 10^{-3}$. Beyond the table: phase 4's Monte Carlo bands use 100 seeds from base 1000 and its saturation diagnostic 50 seeds from base 5000 at $T = 30000$; section 4.4 details the seed conventions. The phase 7 column documents the transaction-cost extension, which was attempted but not fully conducted and is left for future work (section 7.3).

Parameter	1	2	3	4	5	6	7
Traders N	200	128	200	200	200	200	200
Periods T	5000	5000	8000	8000	8000	8000	8000
Price impact μ	0.05	0.05	0.05	0.05	0.05	sweep	0.05
Input AR coefficient φ	0.10	0.10	0.25	0.25	0.25	sweep	0.25
News scale σ	0.01	0.01	0.01	0.01	0.01	0.01	0.01
Null-order scale σ_q	1.0	1.0	1.0	1.0	1.0	1.0	1.0
Forecast window w	–	–	50, 100, 250	250	250	sweep	250
Forecast order p	–	–	1	1	1	sweep	1
Risk scale $a\sigma^2$	–	–	–	0.001	0.001	0.001	0.001
Position cap $\bar{q}$	–	–	–	0.05	0.05	0.05	0.05
Metric window W	–	1000	1000	1000	1000	1000	1000
Diffusion rate π	–	–	–	$0, 3\times10^{-4}, 10^{-3}$	10^{-3}	3×10^{-4}	3×10^{-4}
Exit probability δ	–	–	–	0	0	0	0
Adoption release period	–	–	–	1250	1250	$w+2500$	1250
CE switching (W_A, a, α, β)	–	–	–	–	$(500, 1, -5, 10^4)$	–	–
Cost levels c	–	–	–	–	–	–	8 values, 0 to 0.01
Seeds	42	0 to 99	43	71	71	2000 base, 10/cell	4000 base, 50